in this issue ~
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ESSAY
That Interactive Thing You Do 3 MESSAGE FROM NACIS'
Mic/we/ P. Peterso11 PRESIDENT
FEATURED ARTICLES It is a pleasure to serve as the 1997-
Beyond Graduated Circles: Varied Point Symbols for 6 98 president of NACIS. No, really,
Representing Quantitative Data on Maps I am serious! (as Dave Barry would
Cy11thia A. Brewer and A11drew /. Campbell say). After well over a decade of
association with the organization,
Animation-Based Map Design: The Visual Effects of 26
serving now and then as a member
Interpolation on the Appearance of Three-Dimensional Surfaces
of the board of directors or the CP
Steplre11 Lm1i11, So11ja Ross11111, a11d Slraw11 R. Slade
editorial board, I finally have the
pleasure of serving as NACIS' vice
Decision-Making with Conflicting Cartographic Information: 35 president and now
president. The
The Case of Groundwater Vulnerability Maps
words "pleasure" and "president"
Clwr/es P. Rader and fa 111es O. f anke
are seldom found together in the
same sentence; let me tell you why
REVIEWS I use them together.
The Dartmouth Atlas of Health Care 48
For one thing, I get to deliver
Russell 5. Kirby
some good news to NACIS mem
bers and others with links to
Mapping an Empire: The Geographical Construction of 49 cartography: we have a new
editor
British India, 1765-1843
for Cartogrnplric PcrspectitJes. After a
W. Eliznbetli fepson
two-year search and several guest
editors, we found our new editor,
Atlas of Oregon Wildlife: Distribution, Habitat, and Natural 51 Dr. Michael Peterson of the
Uni
History versity of Nebraska-Omaha, right
James E. Meacl1m11 at the-head of our own editorial
CARTOGRAPHIC TECHNIQUES 53 (conti1111ed Oil page 61)

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NEW NACIS WEB SITE
NACISNEWS 59
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The cover was designed by James Swanson using Adobe Photoshop on a Power Macintosh. James is currently finishing his master's degree at the George Mason University University of Nebraska at Omaha. His thesis concerns the cartographic applications of the Virtual Reality Modeling Language (VRML). VRML is
Jeremy Crampton Judy Olson
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Borden Dent Jeffrey Patton
Georgia State University Univ. of N. Carolina -Greensboro

the Internet standard for the description
of interactive 30 objects and scenes.
Scott Freundschuh Joe Poracsky Across the top of the cover, a series
Univ. of Minnesota -Duluth Portland State University of prism maps show the change in US
income per capita from 1948 to 1970. In
the lower-right is a wire-frame of the
Melissa Lamont Ren Va siliev
extruded states from the VRML map.
Woods Hole Oceanographic State Univ. of New York College The stripe down the middle of
the page
Institution at Geneseo represents a binary data stream with
ones and zeroes. Towards the bottom
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I left is a portion of the source code for the
Matthew McGranagha11 Carolyn Weiss VRML map. Th.is VRML 'world' can be
University of Hawaii -Manoa Statistics Canada accessed at
----- http:/ / osprey.itd.sterling.com I vrml. J
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essay
That Interactive Thing You Do
The incorporation of interaction in the display of maps may be viewed Michael P Peterson
as a major accomplishment of the computer-era in cartography.
Dept. of Geography/Geology
Certainly, interaction has pervaded all forms of mapping, whether it is
Univ. of Nebraska at Omaha
with a database of a geographic information system, a multimedia atlas on
a CD-RO:vt:, or street maps on the World Wide Web. A characteristic Omaha, NE 68182
shared by all of these forms of mapping is the control that the user has
over the resultant map.
However, interactivity is not new in cartography. In fact, it may be as
old as cartography itself. While \.Ve don't know when the first map was
made, it was very likely a product of an interaction between two individu
als. It may have been much like the "paper-napkin map" of today-the
kind of map that is drawn on any piece of scrap paper when words fail as
one tries to explain where something is located. A common characteristic
of these maps is that the person for whom the map is being made will ask
questions that affect how the map is drawn. For example, the person
might ask where a particular landmark is located to provide a point of
reference. The map becomes a product of interaction when the maker of
the map includes the landmark. It is likely that the first map was a product
of this type of interaction.
If the first maps were interactive, what does this mean about cartogra

phy now? One way to answer this is to view the progression of cartogra phy in three stages. The initial maps were interactive, perhaps drawn in sand \Vith a stick. A major shift occurred long ago as a more stable me dium was used and maps were transformed into static objects, first on clay and later on paper. This was an important transition because it made the communication of spatial information possible without the mapmaker having to be present. However, it removed the interactive component. With the help of GIS, multimedia and the web, cartography, now in its third stage, is becoming interactive again.

The first and third stages are both interactive, but differ in how the interaction is achieved - human vs. computer. What do we call the inter vening period? In geology, the time period in which we live is commonly referred to as an "interglacial," i.e., the time between major glacial events. In a similar sense, cartography may be seen as having been in an "interinteractive" period - a period when static maps were the norm. The ice is still melting from this period.

The transition to interactive maps is a difficult one for those accus tomed to static representations. A "paper-thinking" envelopes us. It is a type of thinking that is difficult to overcome because we have been influenced by static maps for such a long period. It is the way we think maps should be. It is the way we have come to know the world.

We should remember that while we have adapted to static maps, many people have not. A large percentage of the population cannot effectively use these maps. They apparently do not find that the information is presented in a usable form. The one million daily hits on interactive mapping services like MapQuest are an indication that interactive maps are much more accessible to many people. While we might criticize the poor graphic quality of these maps, people seem to use them.

How can interactive maps be made better? A conversation is a good metaphor to use in understanding what interaction means with a map. An important aspect of a good conversation is that each participant responds in some way to what the other has said. A bad conversation is character ized by no response, a response that has nothing to do with the topic of the conversation, or does not relate to the previous point that was made.

An additional and important aspect of a conversation is that each participant is building up a database of what the other has said. The database can be fairly sophisticated at times. For example, I can remember stories that people have told me several years ago (although they have sometimes forgotten that they have already told me these stories -a flaw in their 'who-have-I-told-this-to' database). We have not reached this stage of sophistication in interactive cartography. The system rarely remembers what it has already shown. (The closest we get to this is the

short-term caching structure of a World Wide Web browser.)
Viewed in the perspective of a conversation, the type of interaction that we have with maps on the computer is simplistic. It is somewhat like talking to a person for the first time over and over again. There is no database of the interaction. The computer doesn't remember anything. Because it doesn't remember, it cannot raise the sophistication of the interaction.
Perhaps the conversational form of interaction with maps is not we want. Imagine if the computer responded with messages like:
• I've made this map for you before!
• Don't you remember where that feature is located!
• Can we move to a more intelligent level of interaction, please?
We apparently prefer a more shallovv form of interaction with the computer. Perhaps we want to maintain the status of a master. We cer tainly don't want the computer to challenge us as another person might do.
Some computer games incorporate varying levels of sophistication. They adapt to a particular user and move them forward to more compli cated tasks with rewards along the way. This, and the very high level of interaction that characterizes these games, maintain the interest of the user. The user is made to feel part of the game. The interactive map can create the same feeling on the part of the user.
How far we have come from that first interactive map? It might be said that with all of our computers and technology, we still cannot simulate the interaction in a "paper-napkin" map. We may be making more accurate maps and maps that are the result of greater analytical thought, but they may not be as interactive as that first map in the sand.
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The NACIS Program Committee invites you to participate in this meeting by presenting a paper, poster or exhibit. Plan to organize a session, panel discussion, or workshop. A full range of computing facilities at the University of Wisconsin-Milwaukee will be available

for workshops. We look. forward to featuring the current challenges
confronting map librarians and cartographers.
Potential topics include:
handling digital map-library data
-7
compiling metadata
-7
managing the practical constraints of budgets
-7
-7 exploring data using visualization & virtual
XVIII reality
integrating diverse data sources for map
ANNUAL MEETING -7
production
marketing cartography services
-7
developing educational materials
-7
We also encourage presentations that will complement the sympo
NORTH
sium Maps a11d Mi11ds: A History of Cartography i11 Geography
Ed11catio11, which has been organized by the Hermon Dunlap Smith
Center for the History of Cartography, :\lewberry Library. This
AMERICAN
symposium is co-sponsored by NACIS and will be held in
Milwaukee on Wednesday, October 7. To continue themes from the
CARTOGRAPHIC symposium at l'\ACIS, we welcome presentations on maps in
education and on cartographic education from historical, current
and speculative perspectives. Challenges that reflect the practical
INFORMATION side of this topic include production of maps for educational media
and collaboration with authors. Topics on maps and society will
also mesh with the symposium themes.
SOCIETY
Email your 250-word abstract or proposal to the program chair,
Cynthia Brewer, by May 1. List names and affiliations of all authors,
and include phone and email for the presenting author. We encour
age student participation.
Email proposals and abstracts to:
7-10, 1998 cb rewer@essc.ps u.edu
OCTOBER
The deadline is May 1 (Space may still be
available after the deadline).
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Participants will be notified by June 15, 1998,
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of acceptance of their abstract or proposal.

Beyond Graduated Circles: Varied Point Symbols for Representing Quantitative Data on Maps

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Graduated point symbols are viewed as an appropriate choice for many thematic maps of data associated with point locations. Areal quantitative data, reported by such enumeration units as countries, are frequently presented with choropleth maps but are also well suited to point symbol representations. Our objective is to provide an ordered set of examples of the many point-symbol forms used on maps by showing symbols with linear, areal, and volumetric scaling on repeated small maps of the same data set. Bivariate point symbols are also demonstrated with emphasis on the distinction between symbols appropriate for comparison (separate symbols) and those appropriate for proportional relationships (segmented symbols). In this paper, the variety of point symbol use is described, organized, and encourage, as is research on these varied symbols and their multivariate forms.

INTRODUCTION M uch cartographic research has been conducted on apparent-value scaling and perceptions of graduated point symbols. However, discussion of practical subjective elements of point symbol design and construction is limited, especially for the wide variety of point symbols for representing multivariate relationships that are now feasible with modern computing and output. The primary purpose of the paper is to suggest innovative graduated symbol designs and to examine useful combinations for comparing variables and representing proportional relationships. This paper was inspired by the bivariate symbol designs of hundreds of students that have taken introductory cartography courses with Cindy Brewer (at San Diego State and Penn State). These students have been creative and thoughtful in the many ways they have found to represent multivariate map data, with few choosing the old standby of graduated circles when given the flexible design capabilities of the illustration software.

". .. graduated point symbols Although thematic atlases have long made use of a wide variety of

were the most co111111on point symbols, cartography textbooks emphasize use of graduated circles.
quantitative sy111bolizntio11 Mersey (1996) surveyed the qualitative and quantitative point symbols
used in eight recently published atlases and found that graduated point
choice."
symbols were the most common quantitative symbolization choice (36
percent; \·vi th 59 percent of 36 being circles or sectored circles). In compari
son, choropleth maps were used for 24 percent of the quantitative repre
sentations. Mersey notes that, despite the prevalence of point symbol
mapping, choropleth maps are the mainstay of most microcomputer
bascd GIS programs. All of the 13 she reviewed offered choropleth map
ping, with four of the 13 not able to produce graduated symbol representa
tions. Muehrcke (1996), citing Abler (1987), refers to choropleth mapping
as "a cartographic abomination that GIS will sv:iftly kill off" (p. 272) and
then bemoans its continued use. Although GIS has enabled production of
more sophisticated representations, it has also made it easier (for far more
people) to produce choropleth maps as well.
Cartograms and dasymetric mapping offer alternatives to choropleth
mapping, but here the more popular point symbol representations will be

examined. Point symbols are appropriate for quantitative data at point
locations and for areal units. They are not affected by the physical size of
enumeration units (large units produce large symbols on choropleth
maps) and can be used to illustrate a spatial distribution while maintain
ing recognizable geographic boundaries (unlike contiguous cartograms).
Widely spaced point symbols in regions of large enumeration units (such
as counties of the western U.S.) produce the appropriate sense of sparse
data and they contrast well with dense clusters of symbols in areas of
small units (such as eastern U.S. counties). Thus, point symbols preserve
the spatial structure of the entire map without the problems associated
vv·ith choropleth maps or cartograms.
The paper begins with a review of literature on graduated symbols to
update the reader on this long research tradition in cartography and
psychology. The remainder of the paper is devoted to a description of the
v<iriety of symbol designs possible and the nuances in symbol designs that
make them appropriate for different types of bi\'ariate map data. A wide
\'ariety of symbols are illustrated with the hope that these examples •.vill
inspire and encourage the authors of maps and GIS displavs to produce a
wider ,·ariety of engaging and enlightening data displays well-suited to
communicating and exploring their geographic information.
Psychopl1ysicnl Study of Grad1wted Circles LITERATURE REVIEW
Apparent-\·alue scaling and graduated circles ha,·e received a great deal of

attention from cartographic researchers. The circle was, and still is, the most widely used graduated point symbol. Flannery's 1956 and 1971 '"·ork firmly established apparent-value scaling as a topic of research and discussion for the cartographic community.

Co' (1976) concluded that apparent \'alue sc,1ling " ... turned out to be inadequate as a remedy for the underestimation of symbol size ratios" (p. 73). His research examined legend construction and anchor effects associ ated with circle and square size estimation. He found that using a legend containing a small, mid-size, and large symbol (relative to the mapped data) led to the most accurate estimates. His findings supported Dobson (197.+), who stated "a review of psychophysical research on proportional "Appnrrnt-value scnli11g and

circle symbols indicates that readers can interpolate between circle sizes graduated circles /rnpe received a

but that extrapolation is quite difficult" (p. 53). Chang (!977) confirmed great deal of atte11tio11 fro111

Co,'s findings with the results of a similar study comparing size estima tion using ratio and magnitude estimation methods. cartographic researchers."

Macmillan et al. (197-1), Teghtsoonian (1965), Maddock and Crassini (1980), and Shortridge and Welch (1980) examined the effects of the specific wording of the request that subjects estimate circle magnitudes. The latter three e'periments were not entirely applicable to cartography because the standard for comparison needed to be remembered when estimations were made; \·vhereas normally the "standard'' would be simultaneously available in a map legend . .\.1acmillan et al. tested the use of instructions to subjects with a standard present during estimations. They found that instructions for use of a fitting strategy improved the accuracy of subjects' size judgments.

By the mid-1980s, authors became increasingly skeptical about the usefulness of apparent-value scaling and shifted research away from correction by exponent. Griffin (1985) was particularly critical of its usefulness, stating:

Perceptual rescaling may ha,·e merits for particular types of map user, or for the generation of immediate visual impressions, though both cases remain to be proven. Such rescaling pl<1ces increased demands on

map space and, in the experimental context, promotes an increase in the variability of subject responses. (p. 35)

In analyzing intra-versus inter-subject variation in circle size estima tion, Griffin pointed out the ineffectiveness of apparent-value scaling for increasing the estimation accuracy of subjects who performed poorly. He also noted that apparent-value scaling impaired those subjects whose estimates were most accurate. In contrast to Griffin's results, Olson's

(1975) earlier work with circle-size and dot-density estimation examined the interaction between apparent-value scaling and training in the form of practice and feedback about correct answers (for commentary see Will iams, 1977, and Olson, 1978). She concluded that the combination im proves both the accuracy and dispersion of subject estimates, unlike results for either scaling or training alone. Assuming that a map legend functions as training in a limited way by providing example sizes with associated data values, apparent-value scaling may produce improvement in graduated-circle estimation for complete map-reading contexts that is not seen in simpler experimental contexts.

Interactive and animated mapping facilitates examination of multiple symbol scalings. Slocum and Yoder (1996) described student use of Visual Basic to animate series of graduated-circle maps. One approach was to permit interactive selection of scaling values from square-root areal scaling to linear scaling of diameters, which enhanced spatial pattern by greatly exaggerating relative circle areas.

Heino (1991) noted that graded rather than graduated symbols may be the best method of displaying quantitative data on a map. Heino argued that since graduated point symbols were meant as visual abstractions, graded symbols sufficiently convey the intended message without creating the problems associated with graduated symbol perception. Dent (1996) also discussed range-graded circles in his textbook. His discussion in cluded Meihoefer's (1969) ten circle sizes that were "consistently discrimi nated by his subjects" (Dent, p. 174) as well as his own untested set of graded circles. Similarly, Monmonier (1977) recommended a regression based method of circle scaling intended to improve visual correlations between maps by essentially imposing a shared minimum and maximum circle size on the data representations of two (or more) maps to be com pared. His approach to scaling permitted accurate value estimations from legend examples, but correct size ratios between symbols were not main tained.

This shift in research away from apparent scaling was supported by authors such as Worth (1989) who discussed, as Flannery had, some of the problems with approaching cartographic research as a pure science. He noted that much cartographic research borrows from psychology because both fields are highly subjective, and stated that, "By their very nature, experimental tests in cartography must involve many subjective decisions, and although we must do our best to apply the scientific method, subjec

". .. 111uch psychophysical study

tivity will always be invoh·ed" (p. 152).

eliminates spatial structllre

Petchenik (1983) offered an in-depth discussion of cartographic research from our perception of thematic

and the problems associated with limiting ourselves to psychophysical figures .. . " study. She explained that much psychophysical study eliminates spatial structure from our perception of thematic figures, and quoted Chang (1980) who stated, "The stimulus-response relationship for circles is fairly complex, and any correction in map design based on one psychophysical study alone is of limited value, especially given the incomparability between the conditions of the experiment and of real map use" (p. 161). There have been calls for more cognitive research to counter the limita-

tions of psychophysical research in cartography (Olson 1979; Fraczek 1984), and Gilmartin (1981a) discussed the inextricable link between cognitive and psychophysical study. Peterson (1985, 1987) examined mental images of maps and comparison of pattterns on point symbol maps, rather than comparison of symbols. MacEachren's How Maps Work (1995) is a current, in-depth analysis of human cognitive processes used to understand map symbols and map patterns.

Beyond Circle Size Estimation

Groop and Cole (1978) studied the relative effectiveness of cut-out versus overlapping graduated circles, since many graduated circle maps include dense clusters of circles. Their analysis showed that estimation errors associated with cut-out circles were significantly greater than those associated \vi th overlapping circles. Cut-out circles are seldom used today because of the ease of constructing overlapping circles with the computer. However, Dent (1996) noted that they "add a three-dimensional (plastic) quality to the map" (p. 177). Groop and Cole also questioned the effects of clustering on the accuracy of size estimation. Gilmartin (198lb) found that circle size estimation was affected by neighboring circles and that the effect can be minimized by including intervening linework such as bound aries separating enumeration units. Dent noted the current lack of a solution to this problem.

Researchers have examined the effects of both lightness and hue on symbol-size estimation. Meihoefer (1973) studied the effect of transparent, gray, and black circles on size estimation with his l 969 set of graded circles. He found no difference in subject perception with these three fill variations. Crawford (1971) concluded that the perception of circle size was the same whether the circles were represented in black or gray. Patton and Slocum (1985) conducted a study to assess the effect that aesthetic use of color had on pattern recall of graduated circles. Lindenberg (1986) examined the effect of color on size estimation for graduated circles. Neither Patton and Slocum nor Lindenberg found differences associated with color, substantiating Meihoefer's and Crawford's studies. In contrast, Williams (1956) found small differences in size estimates between colored symbols (circles, squares, triangles) and black symbols of the same size,

with the largest difference of six percent between equivalent yellow and black symbols.

Griffin (] 990) recognized the increase in cognitive study of graduated symbol use, but suggested continued research using the stimulus-response approach. He investigated visual contrast between graduated circles and their map background, user preference for opaque versus transparent circles, and the effect of varying circle fill color on size estimation. His subjects disliked a white fill most and showed a strong preference for black figures. Preference results for opacity depended on a subject's preference for clarity of the figure or detail of the background. As in Crawford (1971), Meihoefer (1973), Patton and Slocum (1985), and Lindenberg (1986), color variation was shmvn to have no effect on size estimation.

" .. . color vnrintio11 wns shown to have 110 effect 011 size estimation."

Comparisons ofT ypes of Graduated Symbols

Unfortunately, little has been written by cartographers comparing the relative usefulness of different types of graduated point symbols. This lack stems, in part, from the historical difficulty of manually constructing graduated symbols. Graduated circles were versatile and easy to construct through both manual and automated means, and have, therefore, received greater attention. In addition, few researchers have investigated the

representation of comparison and proportional relationships within multivariate graduated symbols.

In 1926, Eells examined "The Relative Merits of Circles and Bars for Representing Component Parts." He noted commonly held criticisms of using circles to illustrate component parts (what we refer to as propor tional relationships) including: difficulty in rapid and accurate estimation of values, inaccuracy of estimation due to their areal nature, and the suggestion that "while their popularity and psychological appeal are grudgingly admitted their use is 'an insult to man's intelligence'" (p. 122). However, Eells' study concluded that circle diagrams divided into compo nent parts were easily and accurately read, that accuracy increased with the number of subdivisions of circles but decreased for subdivided bars, and that the use of circle diagrams was "worthy of encouragement" (p. 132). Croxton and Stryker's 1927 follow-up study supported Eells's findings and suggested that for illustrating 25, 50, and 75 percent relation

"while their popularity and psychological nppeal are grudgingly ndmitted their use is 'rm insult to mnn 's

intelligence'" ships, circles worked significantly better than bars. Croxton and Stein (1932) further examined the accuracy of size estimates using bars, squares, circles, and cubes. They concluded that scaled bars yielded more accurate results compared to area or volume symbols. Furthermore, they found that performance with both circles and squares was better than with cubes. Their study supported the commonly held belief that the fewer dimen sions a graduated symbol possesses, the more accurately its size is esti mated (a general rule strongly supported by Cleveland, 1985, p. 254). Neither Croxton's studies nor Eells' study was performed in a carto graphic context. Their results offered basic comparisons of graduated symbols without the complications of spatially registered data.

Clarke (1959) examined the relative accuracy of size estimation for lines, circles, squares, spheres, and cubes. His data ~ere collected using three symbol sizes in each trial. Subjects were asked to estimate the sizes of the small and large symbols using a middle-size symbol. Greater errors in estimations of size occurred as the number of dimensions increased and as the difference between the symbol and the standard increased. Ekman and Junge (1961) elaborated on this conclusion by generating power functions for symbol types.

Flannery (1971) included examination of the relative effectiveness of wedges and bars compared to circles in his research. Although he con cluded that wedges were not estimated as accurately as circles, he noted their usefulness for showing proportions for specific locations (cities, ports, and intersections, for example) because the vertex naturally points to the place to which the value belongs. He also noted the relatively high accuracy of estimations with linearly scaled bars.

Crawford (1973) examined perceptions of graduated squares. Specifi cally, he studied the potential for linear rather than areal estimation of square size and whether squares were correctly, under-, or over-estimated. His regression model clearly indicated that square sizes were estimated a really, not linearly, and he showed that square sizes were more accurately estimated than circles. Likewise, Heino (1991) recommended the use of squares and cubes, instead of circles and spheres, for accurate estimation of data sets with large ranges.

In addition to comparing types of symbols, multivariate symbols would seem a logical topic of research in thematic cartography. Both MacEachren (1995) and Nelson and Gilmartin (1996), however, note that very little of this research has been done. One theme in their reviews of the topic is the link between the design of multivariate symbols and whether symbol dimensions representing each variable remain separable or are combined in a more holistic symbol with i11tegrnl dimensions (Shortridge 1982). Nelson

and Gilmartin examine symbols representing sets of four variables with

attention to the multiple purposes of multivariate map symbols: from 'what' and 'how much' at the local level (which may be better represented with separable dimensions) to regional patterns and correlations between variables (which may be better represented with integral dimensions).

As an example of multivariate symbol research, Slocum (1981) exam ined two-sectored pie graphs. Specifically, he measured the just-noticeable difference for sector size and accuracy of sector size estimation. He found that subjects could not discriminate between sector size differences of less than nine degrees, and that sector sizes were estimated within a three percent margin of error. He suggested rounding data to the nearest five percent, rather than one percent, before drawing sectors. His study was, in part, a response to Balogun (1978), who suggested the use of decagraphs (ten-sided polygons) for representing proportions. Examples of decagraph use can be found in the Atlas c~f Ncufo1111dla11d a11d La/1rador, ·where they are used to represent economic data (McManus et al. 1991, plates 10 and 19).

Point Symbol Discussion in Cartographic Textbooks

Cnrtosrnphy: Tlre111atic Map Design by Dent (1996) and Elements of Cartog raphy by Robinson et al. (1995) are two widely used collegiate texts. Chapter 8 of Dent's book offers a concise survey of the research that has been conducted over the past twenty-fi,·e years, including apparent .scaling (Flannery 1971), effects of symbol clustering on size perception (Gilmartin L98lb), anchor effects (Cox 1976), range grading (Meihoefer 1969), and open and cut-out circles (Groop and Cole J978). Dent also directs attention to multivariate graduated point symbols, the use of graduated ~qua res and triangles, and the use of volumetric symbols. Robinson et al. briefly discuss the use of variations in symbol shape and orientation to represent classes of graded data. They offer an in-depth discussion of graduated circle use in representing univariate data with the addition of a color sequence for a second map variable. Their coverage of multivariate mapping is divided into four types, in their words: superim position of features (different symbols), segmented symbols (sectored pie graphs), cross-variable mapping (bivariate choropleth), and composite indexes (cartographic modeling). They warn students that "supermaps" illustrating too many variables often hinder the clear communication intended by the mapmaker.

Issues for Future Research

Point symbol maps may be designed with multiple objectives, such as encourging accurate symbol-size estimation, easing legend matching, "Further experimental testing
attracting attention, representing patterns across a map, and showing
and other i11vestigatio11s s'1011ld
spatial relationships between ,·ariables. Success for one design objective
be structured to account for the

often necessitates failure for competing objectives (Petchenik 1983),
though interacti,·e environments that encourage multiple representations multiple tasks for which maps
improve on this discouraging reality. Further experimental testing and are designed."
other investigations should be structured to account for the multiple tasks
for which maps are designed.
Examples of competing objectives can be found for varied aspects of
point symbol mapping, such as range grading, symbol dimensionality,
map comparison, and multivariate symbols. The literature reveals a
tension between accurate estimation of indi\'idual svmbols and the
alternati\·e of range-graded symbols, for which data values are classed and
assigned to ordered but arbitrarily scaled symbol sizes. Range-graded
symbols are selected to be obviously different in size and easily identified
in a legend. A hybrid approach with symbols scaled to means of data

classes, for example, may be a useful compromise between the comfort of
easy identification and an accurate overview of relative magnitudes across
a map. Similarly, low symbol dimensionality improves symbol-size
"A hybrid approach with estimation but may hinder map pattern interpretation. The larger range in
symbols scaled to means of data sizes of linearly-scaled symbols (bars, for example) causes symbols to
extend farther across the map than higher-dimension and more compact
classes, for example, 111ay be a
symbol forms (areal and volumetric symbols such as squares or cubes)
useful compromise ... "
representing the same data. Adjustments within a map series, such as
equalizing symbol areas among maps before comparison (Peterson 1985)
or using constant symbol sizes for minimum and maximum data values
(Monmonier 1977), assist pattern comparison but interfere with compari
son of data magnitudes between maps. Another challenge is choice of
multivariate mappings with separable symbol dimensions (which may
encourage comparisons within symbols and within variables) versus
integral dimensions (which may improve understanding of patterns in
correlations between variables). Although researching the endless variety
of multivariate point symbols is a daunting prospect, it would be useful to
see work on how people use and understand these types of representa
tions.
The next sections summarize many possibilities for point symbol
mapping. In addition to the theoretical research issues that are reviewed,
interest in these symbols also invites practical research on how point
symbols can be effectively implemented in mapping, GIS, and visualiza
tion software environments.

GRADUATED POINT SYMBOL Figures 2 through 8 illustrate both common and innovative ways to
EXAMPLES AND SOURCES OF illustrate quantitative univariate and bivariate data with graduated point
INI\'OV A TION symbols, and Figure lb presents a choropleth version of the map for comparison. In addition to univariate representations, this discussion concentrates on basic ways of representing proportional and comparison relationships on bivariate maps. These terms require some explanation. 'Proportional' data refers to relationships in which one data set is part of the other data set being mapped. Similarly, Eells (1926) referred to repre senting proportional relationships as illustrating component parts. 'Com parison' data refers to those data that are two separate measures but the relationship between them (correlation) is of interest. One data set is not a part of the other. The data on loan disbursements that was used for the maps in this paper, for example, may be divided into public and private proportions or it may be co111pared to principal repayments for the same year.

Our mapped data (Table 1) were taken from a World Bank report (1994). The same data are mapped in each of the figures to aid comparison of the symbol types, though the small extent of the maps does not foster regional comparisons and does not allow evaluation of the symbols with more numerous enumeration units. All data are for countries in northern South America. Point symbols are used for country data in these maps to emphasize that point symbols are appropriate for areal data, even though some authors restrict their representations to choropleth mapping (Figure 1 b). All data are for 1993 debt levels: total loan disbursements, principal repayments, and the proportions of total disbursements for public versus private loans. Table 1 notes provide further explanation of the variables.

Univariate Data: Total 1993 Disbursements

Figures 1 through 4 illustrate a variety of methods for mapping univariate data. Figure la lists disbursement values as an areal table. Figure lb is a choropleth map of these data, which is shown for compari-

a. Areal Table b. Choropleth

1993

VENEZUELA

Disbursements

2137

D

GUYANA 69

69 Disbursements •

mapped are the

COLOMBIA

amount of each
1697 country's 1993 debt 662
•
flow that is drawn on
loan commitments.
ECUADOR Reported in millions 1551 -2137
662 I BRAZIL of U.S. dollars •
12195
~ Source: 12195
PERU World Bank 1994
1551
c.Shaded
• Squares
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representation-; of 111,111 di'iburscment dat,1. The• many point symbol'> • =:J =1
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in a and b. Number.., (,1) d11 not provide ,1 vi'>ual rL'prL' ...e ntation of
rel a ti\ e magn1tudt'' 11t data\ alues. Thl' ..,1/e' ot the gray fil], of thl' • 662
choropleth m.1p (b) ilrl' controlled by the si/C'i nf the· cnuntril''> ratl1L'r •
th,111 the data\ alu.~'· Cnuntr~ siLl' ha!-> a large visual impact nn tlw
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admoni!->h ,1g.1in<.1 rq>rc,enting total value~ with a choroplcth map. • 12195
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Ecuador 66:2 488 .+97
Guvana 69 46 69
Peru 1551 1007 1.+92
Venezuela 2137 1515 1877
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good,, or 'L'n ILL'' 111 tllL' yc'M 'PL'Cihl'd .. (p. \i'). f'rincip.11 rc•p,11·ml'llh are compared 111
loan d1sbur,c•mc•nh 111 Figure' 6 ,111d 7,i.
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e. Grid Squares f. Graduated

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F1s11rc 2. Areal univariate point symbols representing 1993 loan disbursement data: graduated numbers (a), graduated circles (b), graduated squares (c), graduated ellipses (d), grid squares (e), and graduated pictograms (f). Data values are in millions of U.S. dollars, except for Figure 2a which shows billions of U.S. dollars (see Table 1 for description of mapped data). Symbols arc sized to data values using square-root son but is not a recommended representation for these total dollar values. Figure 1c is a variation on the gray-scale representation of the choropleth map, with the gray fills applied to a repeated square symbol of constant size. This variation removes the unwanted effects of variation in enumera tion-unit size, though differences in lightness are more difficult to compare than differences in symbol area (Cleveland 1985, p. 254). Pazner (1997) proposed a similar representation with constant-size squares that he termed "tile maps." Size and lightness may also be used as redundant symbolizations (each representing the same data values within symbols), and lightness may be used to map a second variable within a graduated

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.:. r r- r: 100 500 5000 by triangles with a constant base and height5 lincarlv scaled to
-
-;'",-,> data (b); and st;ick' of repeated pictogram~ (c).
symbol. These valid variations are not pursued because this paper has
been limited to a manageable set of possibilities by focusing primarily on
graduated symbol forms.
Like Figure la, Figure 2a is an areal table with the siLes of numbers
graduated by their values. Graduated numbers offer the reader the accu
racy of a table with the advantage of a visual representation of relative "Graduated
m11nbers offer tile
magnitudes. (T\'ote that Figure 2a is the only map with symbols that reader tile accuracy of a table
represent debt in billions of dollars; Figure la and all other map legends
with the advantage of a 7.'isual
list debt in millions of dollars.)

represe11tatio11 of relative
Traditional graduated circles and squares also work well with the
disbursements data set (Figure 2b and c). Their familiarity and versatility 111ng11 i
t11des."
make them mapping options that should not be overlooked. By sizing the
height and width of the entire set of symbols by a constant ratio, gradu
ated ellipses (Figure 2d) offer a mapping solution for sets of enumeration

units having generally elongated shapes. These
a. Graduated
ellipses are also included for comparison to a later
Spheres
bivariate use (Figure 6g). Magnitudes represented
with a square grid (Figure 2e) may aid accurate
100
interpretation because individual cells can be
counted.
500 Another areal symbol, the 'money bag' picto
gram (Figure 2f) offers an example shape that
more literally reflects the data, captures the
reader's attention, and may be scaled to represent
relative quantities. Pictograms are particularly
well suited when 'catching' the reader's eye is a
priority, making them common in magazines,
newspapers, and educational material. The New
b. Graduated Stnte of the World Atlns (Kidron and Segal 1984)
Cubes offers good examples of the unlimited possibilities
of pictogram use. This atlas and later editions (for
100
example, the fifth edition was published in 1995)
use graded pictograms to cleverly illustrate
everything from radio receivers per 1000 people to
worker exploitation rates. An example similar to
that used in Figure 2f can be found in the Atlns of
Newfo1111dln11d n11d Labrador (McManus et al. 1991,
pl. 19). Although pictograms may not be well
suited for accurate magnitude estimation, variety
and ability to match the theme of the data being
mapped make them a worthwhile consideration.
Linearly-scaled symbols (Figure 3) work well
c. Block Piles for these data because of the relatively large
100 enumeration units. The bars in Figure 3a overlap
multiple enumeration units and may be difficult to

associate with the correct location. In contrast, the
larger bases of the spikes in Figure 3b better
associate the symbols with their enumeration
units. Stacks of repeated pictograms (Figure 3c)
offer a linearly-scaled version of pictograms (a
similar pictogram is areally scaled in Figure 2£).
Alternatively, scaling the height of a single tall
pictogram for each enumeration unit is also
appropriate. For mapping phenomena such as
mercury in sediments, makers of The Nntionnl Atlas
of Sweden: The Em>iro11111e11t (Bernes and Grundsten
Fi:;:11n• .f. \'olurnctric univariate point symbols reprc,enting 1993 loan
1992) use linearly-scaled graduated columns set on
disbursement d;it,i: graduated spheres (ii), grnduated cubes (b), and
an oblique perspective projection of Sweden
bl<)CI-. piles (c). Cube-root scaling is used to sLi:c the spheres and
cubes, <111d block piles are constrnctt>d by repeating the smallest cube viewed from the
east. This technique produces an
symbol. interesting map and allows the uncluttered use of
linearly scaled bars that would be stacked atop one
another in a more traditional planimetric view
oriented to the north.
Finally, volumetric symbols such as cubes and
spheres (Figures 4a and b) are used for data with
an extreme range and are relatively easy to con
struct. Spheres can be rendered using a circle with
a radial fill and a shadow (Figure 4a). Cube
construction is more challenging because of
projection options (Mackay 1953; Ekman et al.

1961; Brewer 1982; Dubery and Willats 1983; Figure 5). Block
Orthographic
piles (Figure -le) have long been associated with Raisz's talent
for cartographic symbolization (1939, 19-!8). D 1cm
cm
Co111pnriso11 Dntn: Disb11rse111e11ts n11d Pri11cipal Repnyments
1
The possibilities for comparing data with point symbols on
bi\'CHiate maps are limitless. The focus will be on several
Oblique examples
combinations of circles and squares, as well as bars, ellipses, and
cavalier cabinet
cubes. The large number of arrangement options available using 4/11 1

common graduated circles and squares (side by side or ·H=71
m·crlayed) offer different visual impressions while illustrating
LlJ.
the same data (Figure 6a, b, d, and f). Overlapping the symbols LJ)45
produces a more integral symbol that involves pattern recogni usual usual
tion because differences in relative values produce different
overall symbol forms (compare the mostly black symbol for
Axonometric examples
Colombia, which is the only country for which payments exceed
disbursements, to the others). Combining shapes for two shape and area of orthographic
view maintained
(angles sum to 90)
qualitati\'cly different variables, such as graduated circles and
squares (Figure 6f), increases the variation in overall symbol
form<, but makes relative sizes difficult to compare.
There are many options beyond combinations of circles and
squares for comparing data. Adjacent semicircles (Figure 6c)
70
clearly illustrate differences between the two data sets. How
C\'er, the adjacent straight edges may lead to linear rather than
are,1-based estimates of value differences. Estimates of differ
ences between pairs of linear symbols arc easier to make,
although graduated bars (Figure 6e) can be more 'curnbcrsome'
Isometric
because linear ~ca ling increases the range of the map covered by
all scales same
symbols (square-root scaling for areal symbols and cube-root
scaling for volumetric symbols produce more compact ~hapes). Elaborations
One student (Jeff Erickson) offered a creative combination for
di metric:
pairing graduated cubes (Figure 7a). Cubes may also be placed two scales same
'>ide by side or stacked. An example of using pictograms to
tnmetnc:
illustrate comparison data can be found in Thomas's Tl1ird \Vorld three scales ditterent
Alla~ (199-l). He used a split textbook to illustrate "Gross
primary school enrollment ratio" (p. 59) bet\\'een males and Perspective
females. The left side of the book was linearly scalt.?d to the
one-point
percentage of male enrollment; the right side to female enroll perspective
ment. Elaborations
A
Ellipses (Figure 6g) may be constructed with their axes scaled two-point perspective:
to illustrate different sets of data, in this case disbursement on two vanishing points

LJ
the horizontal axis and principal repi1yments on the vertical. three-point perspective:
three vanishing points
:V1,KEi1chren (1993, p. 90) describes this combination as an
integration of attributes merged into one symbol. He cites
Garner (1976) when stating "that object integration is more £1~111,· .i. A „1mpling ot thl'
m,rny \\'ay' of
dra'' mg thrl'e-d1111l'11,u111,1l ,ymbol,, u'ing at
likely to lead to integral or configural conjunction than will two
ern cubl· to dem<•lhtr.11,• tlw l'lfoct'> of variou'
distinct '>patially contiguous objects (e.g., pc1ired bars on a bar rrn1L'<l1Pn „.,ll'm' (Duben·
,md \\ 11lat' 1981).
chart)" (p. 90). The axes of these ellipses were SCi1lcd using the Tlw cubl· pn•1edi<•11'
~L·1wr,1ll\' 111crea'l' in
snplw.;tic,1l1<lll trom thl' l<'P do\\'n in thi-. figure'.
linear dimcn~ions of the bars in Figure 6e. When ellipses are
highly elongated in one direction or the other, the reader will
rccogni/e '>patial patterns in \·ariablc differences by the orienta
tion of the symbols' major axes. Ellipses with more circular
~hapes repre~ent comparisons for which neither of the mi1pped
values arc significantly larger than the other. Ra!:>e (1987)
included bi\·ariate ellipses in his point-symbol mapping soft
ware.

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a. Adjacent b. Graduated
Graduated Square
Squares Overlay
Pnncipal
Pnncpal Dlsbursemenls Pavments
D sbursements Payments
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soo
11 500
c. Adjacent d. Graduated
Graduated Circle Overlay
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g. Ellipses with
• Figure 6. Areal and linear bivariate point symbols for data compari
-. Graduated son. Related variables, 1993 disbursements and principal repayments,
Axes
are compared on each map. :\ote that payments exceed disburse
ments for Colombia. Thus, a representation of repayments as a
Disbursements
on Horizontal Axis proportion of disbursements (a circle segment for example) would be
quite awkward from both practical and logical perspectives. Example
comparison symbols are: adjacent graduated squares (a), graduated
Principal
square overlay (b), adjacent (areally scaled) graduated semicircles (c),
Payments
on graduated circle overlay (d), adjacent graduated bars (e), graduated
Vertical circle and square overlay (f), and ellipses with graduated axes (g).
Axis Ellipse axes in g are linearly scaled to two data values to produce
both varied shapes and sizes, though symbol areas are not directly
scaled to a mapped variable. The overall look of the overlay symbols
(b, d, and f) change markedly depending on which variable is the

5000
larger of the two. Overlay symbol construction is awkward where
amounts are near equal because symbols are almost the same size but
the slightly smaller one will take visual precedence (note Guyana's
gray symbol).

a. Nested b. Graduated
I
Cubes I Segmented
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''1 Cubes
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D soursements Pl)lmtn!$
~ -
F1.x1m• 7 \\,lumetnc bivariate point .,ymbols for both comparison (il) and proportional (b)
rl'iallonships, The same dat,1 represented in Figure'> 6
and 8 are u.,ed for the cube~ in 7a and b rc~pcctively,
Proportional Data: Public and Private Components of Disbursements
Mapping proportional data differs substantially from mapping com
parison data because the smaller of the two data sets is always part of the
larger set, unlike the disbursement and repayment data used in the
comparison examples (Figure 6 and 7a). In Figures 7b and Sa, c, e, and g,
graduated symbols are scaled to represent total disbursements and are
segmented by the proportions of loans to public and private entities.
Symbols of constant size in Figures Sb, d, and fare segmented to represent
only the proportions of the disbursements' compositions, not the total
disbursement value (as the graduated symbols do) so they are akin to
univariate symbols. In all cases, the black portions of each symbol repre
sent disbursements to public entities and the gray portions to private. For
example, the use of graduated segmented circles (Figure Se) allows the "In contrast,
maintaining a
map reader to compare the total value of disbursements among countries
constant symbol size (Figure Bf)
as well as the proportions of private and public disbursements. In contrast,
focuses the reader's attention
maintaining a constant symbol size (Figure Sf) focuses the reader's atten 011
tion on the proportional relationship. the proportio11al relationship."
Examples of sectored circles and wedges (Fig Se, f. and g) are found in a
variety of sources including Historical Atlas of Cminda, Vol. Jll (Kerr and
Holdsworth 1990), The l\/atio11nl Atlas of Sweden: The E1ri.'iro11111e11t (Bernes
and Grundsten 1992), and Tire MnritilJle Provinces Atlas (McCalla 19SS). The
Historicnl Atlas of Canada makes extensive use of sectored circles, and often
uses the space surrounding the map to allow extreme symbol sizing.

Wedges, often used in The National Atlas of Sweden, are particularly well suited for illustrating quantitative data associated with points like ports or cities (p. 97 for example). Wedges can easily be rotated to fit in where other 'bulky' symbols can not easily be associated with a point location. The graduated wedge radius represents the total symbol size without showing the entire circle from which the segment came (Figure Sg). The wedge's point anchors it to its enumeration unit (the opposite association is used for graduated spikes; Figure 3b). The MnritilJle Prot>i11ces Atlas uses a grouping of four wedges set at ninety degrees, like four flower petals. Each wedge represents the amount of a particular type of overnight accommo dation across the region. The reader may either compare entire symbol structures or individual segments of each symbol.

The segmented bars and squares were constructed with fifty-percent markers (Figure Sa, b, c, and d). Students often included these reference

a. Graduated ,,-./?S· b. Segmented
Segmented ·.. ~___...,__~ Bars
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Percent
of 1993 Loan
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to Private
Debtors
~ % Private
Percent of
% Public ~ ~ 1993 Loan
• Disburse
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c. Graduated d. Segmented
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- -)
/ ~.. I Percent
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e. Graduated f. Segmented
Segmented Circles
Circles
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Percent
500
Private
% Private % Public
- -
Percent
Public
g. Graduated
Wedges
'
Percent Private
' Figure 8. Linear and areal bivariate point symbols for a proportional
' data relationship: public and private disbursements are proportions
•4 of total 1993 disbursements. Example proportion representations are:
graduated segmented bars (a), segmented bars of constant size (b),
• graduated segmented squares {c), segmented squares (d), graduated
:- segmented circles (c), segmented circles (f), and graduated wedges
25
(g). Figures b, d, and f ~how only the proportions of private and
public disbur~ement~ and not the total amounts of disbursements,
'
which are shown in a, c, and e. Figure g shows only the portion of
'
I private disbursements as circle sectors, or wedges, with an indication
of the total-disbursement graduated circles in the square-root scaled
·,
sector radii.

points in their symbol designs, which assists accurate percentage esti mates. The symbol form in Figure 8b is known to statisticians as a framed rectangle (Cleveland 1985, p. 208; Dunn 1987, 1988), and Monmonier (1993, p. 65) describes its usefulness in representing both absolute magni tudes and intensities. Dunn (1987) also suggests scaling the widths of the

rectangles to total data values (to total disbursements for Figure 8b). In a conversation about Figure 8, MacEachren mentioned a successful student project from an ACSM design competition that divided graduated squares into ten-by-ten grids such that each cell represented one percent, and each row ten percent, of the whole for all symbol sizes (rather than using grid cells of a fixed size; a hybrid of Figure 8c and 2e).

The work of students new to cartography has been an inspiration. With modern computing at their disposal and freedom from the conventions of traditional thematic mapping, students are producing a wide variety of creative symbols for mapping data usually symbolized by graduated circles or squares. For example, a recent student assignment with eco nomic data produced creative ideas ranging from a proportionally scaled \ilonopoly game character (with pockets turned out or clutching money bags) to weight scales drawn tipped to illustrate balances between dollar amounts (by Elliott Westerman and Erika Bozza respectively). Are these effective symbols? With so little research into multivariate representations, one can not say, but this type of experimentation can be encouraged until there is e\·idence to the contrarv.

CONCLUSION

The intent of this paper has been to review some of the major topics on which graduated symbol research has been conducted and to demonstrate the wide variety of ways to map univariate data as well as bivariate comparison and proportional relationships. The survey \·vas not exhaus tive, but if it has inspired recall of other symbol forms that have been synthesize missed, then it has been successful in provoking consideration of the wide range of possibilities available •vith point symbols. Modern computer mapping allows mapmakers greater flexibility in designing creative graduated point symbols. This increase in flexibility increases the impor tance of research examining symbol design issues. As this article has illustrated, there are wide-ranging possibilities for applying creative, eye catching symbol designs to summarize and synthesize quantitative spatial distributions.

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/11for111atiou Syste111s 1(4):303-326.
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Animation-Based Map Design: The Visual Effects of Interpolation on the Appearance of Three-Dimensional Surfaces

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Computer animation is a potential aid in map design, because it pro vides a means for quickly reviewing many design alternatives. This research is a conceptual exploration of one aspect of animation-based design: The effect of inverse-distance weighting on the visualization of three-dimensional maps. The primary variable examined is the inversedistance weighting exponent. Changing the exponent in small intervals allows the creation of a series of three-dimensional maps

that
can be assembled, and played back as a frame animation. In this proce
dure, cartographers can view the visual effects of various exponents on
the resulting surfaces. Design-based animations allow mapmakers to
visualize effects of their decisions in advance of map production and to
make more informed design decisions. It is suggested that this method
can be expanded to examine map design for a great many forms of
cartographic symbology. Ideally, automatic frame generation and a
graphic user interface should become an integral part of the develop
ment of these visualizations.

KEYWORDS Animation, cartography, map, interpolation, inverse-distance weighting, map design, three-dimensional maps, visualization.

INTRODUCTION The mapping of continuous geographic phenomena in three dimensions presents a significant challenge to the design-oriented cartographer.
The challenge is a consequence of the variety and interplay of design
decisions which, in various combinations, can radically alter the appear
ance of a map, and because of the greater than usual difficulty map-users
have in interpreting three-dimensional maps (Rowles 1978, Jenks and
Crawford 1971). Continuous spatial data is either measured or derived at
control points, which, to a degree, are representative of the phenomena. In
many cases, mathematical interpolation is then used to create a more
extensive network of data values from which a visual depiction can be
subsequently generated. The look of the surface is heavily dependent on
choices made during interpolation, and these choices are numerous,
including grid size and shape, search methods, and interpolation equa
tions (Golden Software 1990). Depending on the interpolation methods
selected and the parameters used in the interpolation algorithm, the
resulting maps may give very different impressions of a single geographic
data set.
Another set of design decisions are encountered following interpola
tion. These are decisions involving the visual orientation of the three
dimensional surface. These decisions deal more directly with visual map
attributes, and for that reason, are more predictable in their visual mani
festations than those involving interpolation. Figure 1 shows a variety of
options including surface rotation, angle of view from the horizon, and
vertical exaggeration, as well as a number of other controls that affect the
eventual look of the surface. Although these choices can also profoundly
influence map appearance, their effects are somewhat easier to anticipate
than those involving the mathematical equations of interpolation. For

2ao0
Vertical Angle. An angle of 90 degrees Vertical Exaggeration. A value of zero would
Horizontal Rotation. A rotation angle of 360

places the viewer directly above the surface show a totally flat surface. degreesplaces the viewer due east of the surface.
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example, it is easier to envision what \·viii happen to a surface if the angle
of \'iew above the horizon is changed from 30 to 50 degrees than it is to
predict the ,·isual effects of changing the inverse-distance weighting
exponent from one to three. Consequently, this research is concerned with
the more difficult case of visualizing the effects of interpolation decisions.
"A11i111atio11s are potentially
While it is true that contemporary three-dimensional surface-producing
valuable aids in selecting opti
software allows experimentation with the effects of design choices on map
mum wnys of designing three
appearance, this is usually accomplished one map at a time and normally
requires many iterations before a satisfactory map is created. For example,
di111e11sio11al maps, representing
SURFER (Golden Software 1990), a popular isoline and three-dimensional data, a11d
gni11i11g a more com
mapping program for Windows/ DOS-based computers, permits the
plete 1111dersta11ding ofg eo
creation of one surface at a time, and does not give the user the capability
graphic data through explora
to view a number of different surfaces in rapid succession. It is difficult
tion of rnrying interpolation
and time consuming to compare the visual results of design alternatives,
and this hampers the cartographer's ability to identify the most appropri options."
ate selections for representing data and communicating accurate and
useful visualizations of geographic data.
Animations are potentially ,·aluable aids in selecting optimum ways of
designing three-dimensional maps, representing data, and gaining a more
complete understanding of geographic data through exploration of
varying interpolation options. Furthermore, many other types of carto
graphic symbols can be explored using similar methods. Although our

focus is just one aspect of three-dimensional mapping, it is not unreason
able to assume that animation can be used to explore variations in unit
value in dot mapping, or the visual effect of varying isoline intervals, for
example.
Visualization
Geographic visualization is an act of cognition through which mental
images of geographic data which have no visual form are developed
(Peterson 1995, MacEachren 1995). This notion of visualization can be
extended to the realm of computer cartography in which data stored in

computer memory can be displayed visually. Cartographic visualization
"Clearly, ani111atio11 e11ables of geographic data can facilitate examination and explanation of processes
mapmakers a11d map-users alike, which evolve over time, over space, or a combination of both (Campbell
and Egbert 1990). It is also a concept which focuses on the exploration of
to perceive differe11ces which
data rather than data presentation. As such, cartographic visualizations
may otherwise go undetected
created with exploratory data analysis in mind are generated as animation
whe11 viewi11g a series ofs tatic sequences used to identify trends and relationships in the data. This
maps, perhaps even the same identification which may or may not be observed in a non-visual setting,
prompts new understanding of research problems or real world applica
111aps that compose the
tions (MacEachren and Ganter 1990).
animation."
In our study map animation is used to construct a sequence of maps
created by incrementing the weighting exponent in the inverse-distance
equation. Visual differences between adjacent maps in the sequence may
at first seem negligible, but examination of a lengthy animation sequence
reveals pronounced variations as the weighting exponent increases.
Clearly, animation enables mapmakers and map-users alike, to perceive
differences which may otherwise go undetected when viewing a series of
static maps, perhaps even the same maps that compose the animation.
Animation adds a dimension to map interpretation and analysis that the
viewing of individual static maps can not provide, just as the experience of
viewing a motion pictures is quite different from looking at a sequence of
individual stills from the same film.
This study is restricted to developing a case study whereby a sequence
of three-dimensional surfaces is produced in which visual differences
betvveen surfaces are a result of small variations in the weighting value
used in the inverse-distance weighting interpolation algorithm. Addition
ally, the method of control point search in interpolation was varied in one
instance in order to provide a slightly different perspective to animation
"Ani111atio11 adds a dimension
based design in three-dimensional mapping. The sequence of changes in
to map interpretation and
surface smoothness revealed by the animation allows map-designers to
analysis that the viewing of visualize and compare the effects of changing variables in the creation of
individual static maps can not three-dimensional surfaces.

provide, just as the experie11ce of
Continuous Surface Interpolation
viewing a motion pictures is
Interpolation of continuous geographic data for the purpose of three
quite different fro111 looki11g at a
dimensional mapping provides estimates of data values for geographic
sequence of i11dividual stills attributes, such as precipitation and temperature, at
unsampled grid
from the same fi/111." points within the area of existing control points. The rationale
behind all
interpolation is that points close together in space are more likely to have
similar values than points further apart (Monmonier 1982). Rather than
drawing boundaries and generalizing the area around known control
points as homogeneous, interpolation is used to model the expected
variation of values within an area (Burrough 1986). Interpolated values are
then used to generate a visual surface representative of the continuous
nature of the phenomena being mapped.

The challenge of interpolation is to choose a technique which creates a
plausible model of the variation between control points. Selection of the
most appropriate method depends on the type of data used, the amount of
computational time available, and the degree of accuracy required (Lam
1983), although this last factor is difficult to anticipate in advance. A wide
variety of algorithms have been developed for point interpolation. For
example, surfaces constructed by kriging, minimum curvature, and
inverse-distance weighting interpolations are shown in Figure 2. As can
be seen, variations in interpolation method can produce pronounced
visual differences. Kriging is a regional variable theory technique that
assumes an underlying linear variogram, and because of this it can pro
duce an estimate of error associated with an interpolation (Burrough 1986,
Sampson 1975). Minimum curvature keeps the grid surface fixed at known
control points while iteratively applying an algorithm to smooth the grid
ded surface. Its primary benefit is that the resulting surface is constrained
to pass through all control points (Golden Software 1990). Other interpola
tion methods include splines, polynomials, Fourier Series, Power Series
Trend, and distance-weighted least-squares (Lam 1983). Each technique
hils a different way of modeling the variation between control points and
produces a map surface with a distinctive appearance. Although it is
beyond the scope of this paper to address benefits and liabilities of each
Inverse Distance Weighting
interpolation method, consideration of these factors is an im-portant =
(weighting 2)
cartographic endeavor, and the reader is referred to an excellent summary

of interpolation methods written by Burroughs (1986, pp. 147-166). Inverse-distance weighting, the interpolation model used herein, was selected for several reasons. Inverse-distance weighting is a relatively simple technique, and due to its simplicity and availability in existing computer mapping software, is commonly used. Widespread application of this technique in cartography makes it a reasonable selection for demon strating its visual manifestations. The general advantages of using inverse di~tance weighting are that computation time is minimal, and the algo rithm is relatively easy to understand (Burrough 1986). This last point will be discussed later in the section of this paper dealing with the detailed interpretation of changes seen in the maps due to variations in the interpo lation weighting.

Most commonly, inverse-distance weighting is a type of moving

Kriging

average computation in which intermediate point values are calculated as an average \'alue of a local neighborhood or 'windm·v' (Burrough 1986). Thb aspect is also referred to as 'piecewise' because only a portion of the control points are used to calculate grid values, although all control points arc c;ometimes used in interpolation (Lam 1983). By limiting the number of control points selected, local anomalies of the data can be represented without affecting the interpolation of other points on the surface (Burrough 1986). However, an inverse-distance weighting technique that encompasses all control points can be used to create a more generalized and smoother surface.

Because observations close together tend to be more alike than distant observations, distance between control points is a key element in the model. The contribution of each control point to the local average is based on its distance from the grid location. Distance is incorporated into the inn~rsc-distance \Veighting equation as the quantity d, as shown in Minimum Curvature cquntion 1. The equation calculates a new data value (Z11ew) for each grid intersection (J) on the interpolated surface. The last variable to be dis

Fig11l"I' .:?. Fn1m)'i<'s of" rnr111/1<111:< 111 tlirce

cussed, Jnd the most important to this research because its visual effect

di111c11,1c•1111/ 'ill/rl<"<'S rcs11/f111g frp111 f/1rec

can be observed in the animations produced, is the weighting exponent diftl-r<'lll

111lt'r/l<'illlio11 111cflw1I-. 011111 llllllJ'pt•d

(1ul J. As the weighting exponent increases, the influence of nearby points 11re f(ltal

/11/11. /995)'r1•Lip1tal1c>11.f(11·1\'ebmskr1.

I (Zi/d/vt) where:

Z = z value

I: (l/d.wt) d =distance

wt = weighting value

l
Equation 1. /11verse Distance Weighting Eq11atio11
on the local average also increases while the influence of more distant
points decreases. Emphasizing nearby points creates more peaks and
valleys in the surface in contrast to the smoother appearance resulting
from the use of a low weighting exponent which gives distant points more
numerical equity in the computation.
A single set of data values located at control points can be subjected to
inverse-distance weighting interpolation and then, by incrementally in
creasing (or decreasing) the weighting exponent in small numerical steps,
a sequence of maps showing weighting exponent effects on visual surfaces
can be generated. Maps thus generated can be played back as a frame-by
"Maps thus generated can be
frame animation, and the effects of differences in the weighting exponent
played back as a frame-by-frame
can be observed. This can be done for either isoline or three-dimensional
animation, and the effects of maps, but only three-dimensional maps are used in this research. The
differences in the weighting mapmaker, after a thorough examination of the animation, can then
exponent can be observed." choose the map deemed most appropriate for the mapping purpose at
hand, or the model or theory being examined . This type of visualization
through animation can serve as a map-design tool whose utility extends
well beyond visualization of weighting exponent effects and into numer
ous other arenas of map design, in ways that Peterson first explored in his
sequencing experiments with choropleth maps (Peterson 1993).
Constructing the Animations
Data for the animations was acquired from weather stations across the
state of Nebraska. A total of seventy stations were used. At each of these
control points (locations shown in Figure 3), precipitation for the month of
July, 1995 was recorded. Animation frames were created in SURFER, v 4.5
(Golden Software 1990), and then imported into Animator Pro, v 1.2
(Autodesk, Inc. 1991) for playback. Figure 4 shows a selection of frames
from two of the animations that were eventually constructed.
Control points used in the animation were selected because these
climate stations have fairly uniform spacing throughout the mapped area.
Inverse-distance weighting interpolation is easily affected by non-uniform
• •
3•.14 1•.74
1.9 3.98 1•88
4.•35
1.•47 1.•57 1•.8 2 •61 0•.93

1• 24
2•.95 1•.4 2•.56 1•.4
2•.05
2•.99 2.34
0•87 1.•63 • 2•.28 2 25•
I• 69 1•3 • 8 3•33 4•7 1•46 0•95 1•.331 • .1 2.35 0•.97 I
1.58 I• 53 2•29 2•. 2 2 •. 1 2 2 1.•87 1• 7 2 1 •0 1 0•.8 2•.17 1•.46 • o• .•8 .9 9 0 1 •7 . 4 1 6 •
1 •.0 • 3 2 .26
2•.65 2•.34
2•14 2• 01 2•.53 2 7•5
1•.9• 1•.47 2•.64 335.
2• 13 ·-3• 25
2•6 1 : 1•.59 I 7 3•.n 5.64 •
212•
Figure 3. Location of co11trol points and associated total /11/y, 1995 precipitation values for the maps.

Number 29, Winter 1998 31
WT=1 WT=1
WT=2 WT=2
WT=3 WT=3
WT=4 WT=4
WT=5 WT=5
WT=6 WT=6
METHOD=ALL METHOD=NORMAL
Figure 4. Frames scle«tcd from a11i11111lio11;;. A11i111atio11s /inpr a Iola/ vf120 fr11111e:<. The method
,.de,.red lo as ALL 11'.'cs ali co11trol poi11I «alue,; i11 iim:rs,• disfm1cc <i>;ig1ttii1g. The mctlrvd rcfa,wl to
"One long-standing, useful,
as NOR.lvlAL 11:<es a rcgii•11al selecf1011Lfco11trol17vi11ts based 011 a sclecf1,111of11ear-11eighl>ors. \.\Tis
f/1e weigl1liJ1S <'Xi'OJICl/f. and often violated convention in
graphics and cartography
distributions of control points because equal weights are assigned to each
maintains that the selection of
point even if it is in a cluster (Lam 1983). In reality, the mapmaker does not
symbols for maps and graphs
ahvays have control over the locations of control points, but for the
purposes of this study it \.Vas desirable to avoid the effects of uneven or should reflect the nature of data
clustered arrangements. to be mapped."
One long-standing, useful, and often violated convention in graphics

and cartography maintains that the selection of symbols for maps and graphs should reflect the nature of data to be mapped (Tufte 1983,

Robinson and Petchenik 1976, Dobson 1973). When studying the effects of inverse-distance weighting in three-dimensional animations, continuous data is most appropriate for visualization. Precipitation is a truly continu ous geographic phenomena, and for that reason was chosen for anima tions in this study.

Individual animation frames (maps) were constructed in SURFER by entering control point x, y locations, and corresponding z-values (precipi tation) for the seventy climate stations, and then computing the inverse distance grid interpolation. For each successive frame, the weighting exponent was increased by a value of 0.05. This fairly small increment was chosen to produce a smoother animation, and one which lasted a sufficient length of time to allow examination of the visual effect of increasing weights. An attempt to utilize an increment of 0.1 was made but rejected because the animation appeared too discontinuous and jumpy between successive frames. Weighting exponents ranged from 0 .05 to 6, stopping at 6 because higher weights produced little change in the three-dimen sional surface. This exponent range and numerical increment yielded a total of 120 frames.

Most visual parameters were set and remained constant throughout production of individual frames for each animation. In order to examine the visual effect of other interpolation settings, some of these settings were varied between animations. Uniform vertical scaling was used in all animations, and was selected by first displaying the surface with the greatest vertical relief (wt=6}, adjusting the vertical scalar until no clipping of the surface at the edge of the screen occurred, observing the numerical index of the vertical scalar (in this case, an index of 1.5), and then setting all subsequent frames to that index value. Angle of view from the horizon and horizontal rotation of the surfaces were set, and also remained con stant throughout the animations. Angle of view was set at 45 degrees above the horizon, and horizontal rotation at 235 degrees (0 being a view due west). For the precipitation data used in this study, this orientation was chosen because it was the best for revealing the peaks, valleys, and spatial trends of this rather complex surface, while it also minimized hidden surfaces. Once initial settings were determined, individual frames

"Weigh ting expo11e11 ts greater

were produced and screen-captured to GIF file format at 640 by 480

thn 11 Jou r produced su rfnces that resolution. Frames were then imported into Animator Pro for playback.

appear to be identical. It was Two variations of the animations are shown in Figure 4. Settings, as

described above, are the same for each animation; the only variation is in
also observed that vertical relief
the way control points were searched for and selected for interpolation.
i11creases as ·weighting expo-
METHOD=ALL indicates that all control points were used to derive the
11e11ts increase because increas
interpolated surfaces, but they were still subjected to inverse-distance
ingly greater weights are ap weighting. METHOD= NORMAL indicates that the number of control
plied to fewer control points. points was restricted to a set number of neighboring control points within
a specified radius. One can observe that the animation produced as ALL
This occurs regardless of control
has slightly smoother surfaces when lower weighting exponents are used.
point search and selection
However, the surfaces rapidly become similar as weighting exponents
methods. "
increase. Further experiments with other search settings revealed the same
trend. Weighting exponents greater than four produced surfaces that
appear to be identical. It was also observed that vertical relief increases as
weighting exponents increase because increasingly greater weights are
applied to fewer control points. This occurs regardless of control point
search and selection methods.
These observations are indicative, but not exhaustive of the utility of
this sort of visualization, and the conclusions derived from our observa
tions would have been difficult to arrive at by any other means. Selection
of individual maps to be representative of an animation series will ulti-

mately be determined by the author's purpose and goal, statistical consid
erations, the intended audience, and the author's knowledge of the
geographic phenomena being mapped. In this regard, design-oriented
animations can assist authors in making more informed map selections.
Animation-assisted map design is a promising tool for cartography, and CONCLUSION
can be extended \Nell beyond animation of the visual effect of inverse
distance weighting on three-dimensional mapping. Virtually all variables
encountered in designing and constructing three-dimensional surfaces can
be animated, including data search procedures used in gridding, surface
rotation, vertical scaling, and horizontal angle of view. The greatest
potential for this method lies in examining design decisions whose conse
quences are difficult to visualize mentally, such as those dealing with the
mathematics of grid interpolation. Jn a further example of related uses of
X=20
animation, Figure 5 shows frames from an animation in which only the

density of the interpolated grid is varied. Mapmakers can examine this
animation to determine an appropriate level of grid-based generalization
for the final product. The first frame (top) shm.vs a surface that is made
from a grid that b loo coarse to give an adequate impression of a continu
X=40
ous surface, while the last frame's surface is made from a grid so fine that
line svmbols coalesce and obliterate portions of the surface. The animation
shows that neither of these two choices, nor some of the other frames,
would be appropriate. Hmvever, the actual selection of a surface must be
made in consort with the purpose of the map, the mapmaker's intentions,
and statistical guidelines, such as the error generated by interpolation.
Animation-based design is not limited to examination of three-dimen X=60
sional surfaces, and could just as easily be employed in examining design
variables in isoline, choropleth, dot, and graduated symbol mapping, as
well as other forms of symbolization, (DiBiase et al.1992). fn order to
become a truly useful design tool, creation of frames for animations
should be automated, and the means for selecting variables for animations
should be interactive and employ a graphic user interface (GUI). X=80
Automated map generation is desirable because the construction of
animations manually, one frame at a time, is a costly and time-consuming
process. Unless animations can be produced automatically by interfacing a
geographic data base vvith map construction software, animation-based
design is not likely to be widely employed by cartographers. Availability
X=100
of automated animation would provide a means to quickly construct and
playback animations, make design decisions, and if a single map is de
sired, to produce the most appropriate map for display or publication.
The addition of interactivity and graphic user interfaces to auto
an.imation software gi\·es even greater utility to animation-based design
and e\.ploratory data analysis as well (Peterson 1995, Lodding 1983). The
X=120
interface should permit users to make design decisions by selecting data;
interpolation method; design elements, either singly or in combination, to
be varied through animation; legend structure and other ancillary map
elements; and the number of frames. It should then render the animation
and permit playback. The interface should allow the user to change
animation speed, reverse the animation, pause and resume, and possibly
X=140
to query the numerical geographic data base. Finally, the interface should
permit the user to record their selections and produce a final map. In order
for animation to become a robust means for design-based visualization, it
is essential that these capabilities be repeatable. When this occurs, cartog F(-.:11rc 5.
Frames o:clt'cted from 1111 111111111111011

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raphy will be closer to utility in the use of animation in map design and
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Decision-Making with Conflicting Cartographic Information: The Case of Groundwater Vulnerability Maps

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Conflicting cartographic information can cause problems when used to support planning decisions. Creation of conflicting information is becoming more common as geographic visualization and modeling software are used to develop multiple maps that represent different views of the same data. This paper presents groundwater vulnerability mapping as an example of information conflicts of this type. Three different vulnerability models applied to the same test data produced radically different results. This information was presented to a group of local planners to examine how they would deal with the conflicts. Through this exercise it became apparent that each planner used highly individual criteria to evaluate the results from the models. A continuum of strategies describes the range of responses from aspatial to spatial approaches. Jung's theory of psychological types is applied to further understand variation in responses. A venues for further research are suggested in the representation of cartographic information conflicts, the role of psychological types in decision-making with maps, and the role of group dynamics in decision-making with maps.

INTRODUCTION

Conflicting visual information in cartographic representations is a potential problem in all map displays and geographic analyses. The use of multiple representations for both display and analysis has become common practice with the widespread use of geographic visualization and analysis software (e.g., geographic information systems -GIS). These software products aid in the generation of many different views of one data set, the comparison of different data sets, the use of different methods for processing data, and the construction of different analytical models. The use of these tools for purposes of visualization provides a context through \-which many additional insights about a topic may be gained. In this paper, the question of conflicting information in cartographic displays is pursued in order to assess variations in map use strategies and interpretations when information is contradictory. This issue became apparent during a project that investigated the use of GIS for implement ing three different groundwater vulnerability models from the same data set (Rader and Janke, 1995). Different representations of groundwater vulnerability effectively illustrate the use of multiple views as suggested

by Tufte (1990), Muehrcke (1990), and Monmonier (1991), since maps produced with these models supposedly represent 'similar' information. However, the maps (see Figure 1.) are quite dissimilar, and the conclusions derived from their use may be contradictory. Visual information conflicts such as these may be common to many GIS analyses.

Maps produced from groundwater vulnerability assessments are a primary information source employed by decision makers in developing landuse policy. Decision makers often have neither the original data nor the knowledge to assess the validity of the different models. In many cases, decision makers fail to distinguish between the maps and the models that they represent. With increased use of GIS and modeling techniques, multiple maps representing 'similar' information will likely be available and produce information conflicts for decision makers. Ultimately, knowledge about how people resolve problems with conflicting

"In many cases, decision makers
Jail to distinguish between the
maps and the models that they
represent."

visual information should provide strategies for improving both cartographic displays and users' interpretations of these displays. To examine this question, a map use exercise based on maps from different groundwater vulnerability models was presented to a group of local planners, elected officials, and groundwater specialists.

The first part of this paper describes the vulnerability models, the context for their use, and a more formal problem statement. An overview of the map use exercise and the results are then presented. Finally, issues pertaining to the strategies that decision makers employ for dealing with visual information conflicts are discussed. In approaching this study, it is hoped that concrete recommendations can be provided on how to resolve information conflicts between maps. However, the participants' disparate responses indicated that the fundamental role of maps in decision-making was at issue and that diverse approaches to problems of conflicting information were in operation. Parallels with Jung's psychological typology are drawn upon to help understand variations in decision-making strategies. Finally, avenues for further research are suggested.

Conflicting Visual Information in Cartographic Representation and Planning

The potential for conflicting cartographic representations appears in several ways. The desirability of multiple maps for purposes of visualization has been documented by several authors. Tufte (1990) noted that small multiple maps are well suited for time series data; Monmonier (1991) suggested that multiple views of a geographic data set are more truthful (i.e., ethical) because they provide a comparative frame of refer

ence; and Muehrcke (1990, 9) observed that a more thorough understand
ing may be obtained by using several different maps created from the
same set of data. However, while additional insights may be generated
through multiple views, visual information conflicts may arise if views
display contradictory information. The question that should be asked is,
what happens when map users are presented with conflicting or contra
dictory information?
Little work has addressed the problem of conflicting visual information
between multiple cartographic displays. Typically, cartographic studies
have examined map similarity and pattern comparison (Olson 1972,
Monmonier 1974, Lloyd and Steinke 1976, Peterson 1985, and MacEachren
". .. wlrat happens when map and Ganter 1990). These studies have examined perceived relationships
users are presen fed with con between several maps, impacts of map complexity, and map similarity.
While these have provided a background for understanding the perceptual
flicting or co11 trad ictory infor
characteristics of map displays, the question of conflicts in information
mation?"
content between several maps remains. This becomes increasingly impor
tant in geographic information processing technologies where multiple
displays may be produced from the same data set. For example, different
views may be accomplished through changes in classing and symboliza
tion schemes or the combination of multiple layers in overlay analysis.
Wood (1992, 186) suggests that presenting multiple "relationships
constituted by the interplay of the data" is a desirable artifact of the
mapping process. Muehrcke (1990) has extended this discussion to include
the idea of "map stability." Map stability refers to whether or not changes
in the way data are processed or symbolized have an impact on the
message perceived. However, what happens when views directly conflict
with one another? While seeing small differences in representations
illustrates the impact of cartographic methods on displays, seeing large
differences may call into question both cartographic and modeling meth
ods and possibly their ultimate utility.

Planning involves the design and consideration of a series of alterna
tives. Monmonier (1991) reviewed the use of maps in zoning and environ
mental protection. Ylaps serve as tools for making planning decisions and
communicating a plan's spatial implications. In DiBiase's (1990) con
tinuum of visual thinking and visual communication, planning maps
operate in both the private and public realms. The maps are used in
planning policy decisions, essentially a thinking task, and in presenting
planning policy decisions, essentially a communication task. In this

context, the power of maps to persuade should not be overlooked (Harley 1989, Wood 1992). The 'official' nature of maps often operates at a subconscious level.

Monmonier (1991, 71) notes that "without maps, planning would be chaotic, furthermore even with maps, many would argue that planning is chaotic." In addition, he comments that errors in map compilation can be significant and that the same information is often used to develop different plans (Monmonier 1991). Pla1ming scenarios often involve creating different conceptual models about a future reality. Different scenarios can often produce markedly different results. The 'stability' of results may be re-framed with the concept of 'model stability.' Model stability describes how representations change through the use of different models for

"While individual data layers

combining data layers that were processed using the same cartographic methods. This is a case common to many GIS analyses that employ may luwe high map stability,

different conceptual models for describing a process. While individual the fi11al 111aps made from

data layers may have high map stability, the final maps made from different models may be different models may be unstable. The inherent stability or instability of unstable."

the models may influence how decisions are made with different maps. The implementation of different models therefore has the potential to produce information conflicts. Conflicting information is a major source of chaos in the planning process.

Groundwater Vulnerability: An Example

Maps that result from groundwater vulnerability models provide an apt example of potential cartographic information conflicts in planning. Beginning in the early 1970s, a variety of vulnerability models were developed in response to environmental concerns about drinking water. Significant health effects from drinking contaminated water have been documented; these include cancers (Armijo 1981; Lee and Nielsen 1987; Canter 1987), blue-baby disease (Lukens 1987), and central nervous system birth defects (Scragg et al. 1982). By the mid-1980s, it became apparent that planning and management tools were needed to identify places that have high vulnerability to contamination (Aller et al. l 985).

Several types of models have been developed to assess vulnerability. These models can be grouped into four major categories: hydrogeologic setting models, parameter ·weighting/scoring models, empirical models, and simulation models (Geraghty & Miller, Inc. and ICF, Inc. 1990). Hydrogcologic setting models use qualitative site-based assessments to rank the relative vulnerability of different areas with varying geomorphic, geological, and ground water characteristics. Parameter weighting I scoring

models numerically rank and '"reight susceptibility factors and calculate relative vulnerability scores for areas with different hydrogeologic charac teristics. \IVeights are often assigned to account for variations in the relative importance of factors in particular situations, e.g., areas with thin soil. Empirical models use data that relate knO\·Vn occurrences of contami nants in groundwater to hydrogeologic characteristics through the applica tion of leaching models or statistical inference. Finally, simulation models attempt to predict contaminant leaching through interactions beho\leen

hydrogeologic characteristics and contaminant fate, such as dilution, breakdovvn, absorption, and volatilization (Geraghty & Miller, Inc. and ICF, Inc. 1990, 6-1) .

.\1ost of the models that have been applied in the context of GIS have been hydrogeologic setting or parameter weighting/ scoring models. DRASTIC (Aller et al. 1987) is a general-purpose, parameter weighting model that was developed for the US EPA for vulnerability assessments at county and state levels. Several other models have been developed for state-wide assessments, including the Wisconsin Susceptibility Model (WISM) (Schmidt 1987), which uses weighted hydrogeologic characteris tics to model vulnerability, and an aquifer vulnerability model of Michigan (Lusch et al. 1992), which uses hydrogeologic characteristics for determin ing relative risk. All of these models use a combination of data layers that include soil, subsurface geomorphic characteristics, geology, recharge rates, and topography.

Little research has addressed differences in model performance. The authors of the models often state that results from different models are not directly comparable (Aller et. al 1985; Schmidt 1987). Furthermore, Mer chant (1994) notes that there is little validation of these models in the context of GIS and that there are a number of questions concerning the impact GIS processing methods have on model performance. Rader and Janke (1995) implemented several models popular in Wisconsin to address the question of model differences. In applying the models, the data layers and processing methods were held constant so that variations in the results were due to differences in model design.

Three models were used: SCAM (Soil Contamination Attenuation Model) (Zaporozec 1985), WISM (Wisconsin Susceptibility Model) (Schmidt 1987), and DRASTIC (Aller et al. 1985). All are parameter scor

"The basic problem is that eue11

ing/weighting models that use several data layers, although not all

when using a common data set

models use all of the layers, nor do they consistently weight the layers.

processed in the same Wat/, the Maps derived from the three models for a site in St. Croix County, Wiscon

results are vastly different." sin are displayed in Figure 1. The basic problem is that even when using a
common data set processed in the same way, the results are vastly differ
ent. It is important to reiterate that the models' developers state that the
models are not directly comparable. Regardless, it is likely that the maps
represent the same thing to the lay person and many decision-makers.
Statistical analysis of the relationships between the models further
corroborate the visual disparities (Table 1.). Correlation coefficients
demonstrate little or no relationship in the SCA.\1 -DRASTIC pair and
very weak relationships in DRASTIC-WISM and WISM-SCAM pairs. The
comparison map and coefficients of areal correspondences between the
models indicate complete divergence over 22 percent of the area, with
similar ratings between two of the three models over 70 percent of the
area, and complete correspondence over 7 percent of the area. The coeffi
cient of areal correspondence only indicates agreement or disagreement
across the models. Therefore, the models produce conflicting risk ratings,
both graphically and statistically.
Maps produced from these assessments are often the primary informa
tion used by officials for making land use policy decisions concerning
groundwater. Often, officials neither have the data nor the knowledge to
assess the validity of the models. This has led to the following question:
Given that groundwater vulnerability models produce different results
and, therefore, different maps, how do decision makers respond when
confronted with conflicting cartographic information?

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WISM COMPARISON
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Figure 1. Gro1111d1mfa V11/m•rnbility Model Results
In asking this question, an attempt is made to understand how visual data
impact decisions, especially where multiple views of the data present
conflicting information.
In order to examine the impact that conflicting visual information has on METHODS
planning decisions regarding groundwater protection, the study was de
signed using the three conflicting maps seen earlier in Figure L While the
models are not explicitly designed for site-level assessments, they serve a
function in screening sites for further consideration. In all cases, a site-level
assessment would need to be conducted prior to a final decision. "While the models are not
Thirteen local decision makers participated in the map-use exercise. explicitly designed for
site-level
These people \.Vere chosen because they represent typical users of vulner
assessments, they serve a
ability maps in planning-based decisions. The participants included five
function in screening sites for
planners, three local government board members, two geologists, one civil
engineer, one developer, and one farmer. Only three of the thirteen further consideration."

Correlation Coefficients Coefficient of Areal Correspondence
(standardized unclassed scores) (classed scores)
SCAM-DRASTIC 0.049 None 22.4%
DRASTIC-WISM 0.279 Partial 70.3%
WISM-SCAM 0.311 Complete 7.3%
Ta/1/c 1. Rclatio11slrips /1efwee11 Gro1md1mfer V11l11erability Models
participants had significant experience with groundwater vulnerability
models and mapping.
The exercise had the participants identify potential sites for houses
using the information on groundwater vulnerability for each of three
"The exercise had the partici
different maps. Each map was divided into four quadrants. Participants
pants identify potential sites for were asked to assume that all sites were equally accessible
within each
/rouses using the information on quadrant and that each site would have its own well and
septic system.
grou 11dwater vulnerability for For each map, they also rated the vulnerability for each
quadrant, chose
the best location for a house in each quadrant, and selected the overall best
each of three different maps."

quadrant for a house. A sample test page is reproduced in Figure 2. From these questions, information on perceived relative vulnerability levels and locational behavior were obtained. Such site selection problems are realistic and critical tasks for planners.

The three specific task questions were:

1) Based on this map, rate the average risk for groundwater contami nation in each of the four quadrants using the scales below.

2) Based on the information on the map above, mark the best location for a house in each quadrant with a dot. Assume that all locations are equally accessible and that each house will have its own well and septic system.

3) Examine the map above. Which quadrant contains the most desirable location for a house? You must choose one.

ln designing the test pages, the three maps were simply labeled as Areas 1, 2, and 3. Since the vulnerability maps had extremely low correla tion coefficients (0.049 -0.311), the maps were not rotated or flipped to mask that the maps represented the same area. In general, as the correla tion coefficient of the map pairs falls below 0.84, it becomes increasingly more difficult for subject to judge similarity (Olson 1972). None of the

"Participants were then asked to participants noticed that the maps represented the same area during the

describe /row they would resolzie exercise. This slight deception was later revealed to the participants and

formed the basis for a discussion of how to resolve information conflicts

the i11formation conflicts

between the maps.

betwee11 the maps, considering

Once the participants answered the three sets of questions, they were

that tire maps represented the

given a description of how the different maps were constructed using a

same area. GIS. At this point, the fact that the three maps represented the same area was revealed. Participants were then asked to describe how they would resolve the information conflicts between the maps, considering that the maps represented the same area. After answering this question individu ally, participants were divided into two focus groups for a discussion of the models and an opportunity to examine how information conflicts might be resolved in a group setting. Each of the authors facilitated one of the groups and transcribed the discussion. After the focus groups finished, each participant responded to a questionnaire on their previous experi-

Groundwater Vulnerability Area 3

Ri11k Potcnti&l

- High

- Moderate High
- Moder&Le Low
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WJ>h: 1i)$191TI.
Oue&tion 3 &arNM the map ai)ove. Whrel'I qu:adrw COl\talM the IN>&t des.uable location ! or a l'loU&e?
You !Nllil .moo.a Ol\e. 1 2 3 4
Figure 2. Tc::t l'asefor WISM M,1dclfrom Map Use Exercise
ences with vulnerability maps and models and their uses in landuse
planning.
Analysis of \·ariance (A>JOV A) was used to test the relationship between RESULTS
the vulnerability ratings for each quadrant. The results indicated that the
participants' a\·erage vulnerability ratings in all but one quadrant '"'ere
significantly different (p less than .OS) between most map pairs. (See Table
2.) Quadrant three (SW) was the only quadrant where the ratings were
similar between all maps. Overall vulnerability ratings for each map were
compared using A.t\OV A, and all map pair comparisons were significantly "The results indicated that the
different (p less than .01). Therefore, the participants' perceived vulnerabil participants' average vulner
ity ratings vvere significantly different. This indicates that the maps exhibit
ability ratings in all but one
a potential for information conflicts.
quadrant were significantly
Quadrant four (SE) was judged by the participants to be the most

suitable location for a house on all of the maps. This quadrant also re different (p less than .05)
ceived the lowest average risk ratings for each of the maps. This indicates betiveen most map pairs."
that even though there were significant differences in vulnerability ratings
for this quadrant, it was still the best general area for a house. In this

Quadrant 1 Quadrant2
SCAM-DRASTIC* SCAM-DRASTIC*
DRASTIC-WISM* DRASTIC-WIS M
WISM-SCAM* WISM-SCAM"'
Quadrant 3 Qoadrant4
~
SCAM-DRASTIC SCAM-DRASTIC
DRASTIC-WIS M DRASTIC-WISM*
WI SM-SCAN WIS M-SCAM*
"'SIGNIFICANTLY DIFFERENT at p < .05
Table 2. V11/11aal1i/1t,11 Rati11gs /1ct11>ee11 Map Pairs by Q11adrn11t (ANOVAi
context, the information content was reasonably stable. In other words, the
maps produced the same result for the determination of the best quadrant
for a house although the maps were different.
The best location for a house within each quadrant varied by map
(Figure 3.). It is not surprising that in areas where locational choices are
constrained, more clustered patterns occur. SCAM and WISM models
provided high constraints, and therefore the participants' responses
tended to cluster. These clusters occurred in different locations in all
quadrants except quadrant three (SW). The DRASTIC model, in contrast,
had fewer constraints on location. The patterns produced were dispersed
in the northern quadrants, and as constraints became more severe, the
patterns became more clustered in the southern quadrants. The southern
clusters occurred in different locations on all three maps.
DISCUSSION Differences in both vulnerability ratings and locational preferences indi
cate that different results would be obtained depending on which map
was used for analysis. The problem, as stated, is what happens when
people are confronted with two or more maps that present contradictory
results. It is important to remember that the results produced with the
different models were based upon the same data set, although the models
incorporated the data in different ways.
To examine how people deal with conflicting cartographic information,
the following question was asked:
Given that these results were developed from the same data set, for the
same area, and yield conflicting information, describe how you would
deal with this in the context of making planning decisions?

After writing a response to this question on their own, each participant
joined a focus group for a discussion. This allowed an examination of both
"Some participants suggested individual and group decision-making strategies and to ask additional
throwing out the maps follow-up questions concerning the use of the maps and their utility.
completely and using 'common After examining the individual responses, a number of similarities and
differences in participants' comments were noted. Responses varied by
sense.'"
how important maps were to the decision-making process. Some partici
pants suggested throwing out the maps completely and using 'common
sense.' Others defined the problem as political, referring to administrative
and zoning codes that dictate acceptable procedures and regulations and
require minimal map use for arriving at a decision. In the middle, some
participants suggested using a combination of the models or using the
models as 'advisory to decision-making.' Finally, some participants
suggested 'improving existing models or developing new models' that
would provide more valid maps.

The importance of the map and spatial approaches to the problem DRASTIC
varied greatly. These approaches could be characterized by a continuum
from aspatial to spatial (Figure 4.); with less reliance on the map defining
the aspatial end and more reliance on the map defining the spatial end of
the continuum. The implicit assumption at the aspatial end of the con
tinuum is that the map is unnecessary. Common sense, unreflective
opinions, or rules-of-thumb fall at the aspatial end of the continuum,
because they often do not explicitly consider local ecological conditions.
For example, "everyone knows that you don't put your well next to your
septic system." While initially these may appear to be spatial approaches,
these approaches do not involve the evaluation of specific site characteris
tics. In certain circumstances, for example, in areas with deep wells, the
'rule' may have little practical meaning. The issue is that one does not
need map-based information to apply the rule. The implicit assumption at
O Hou•e locwon (DRASTIC)
the spatial end of the continuum is that a map is needed, but that the
existing models need to be refined.
SCAM
The question arises, why do people view the utility of maps (and
models) with such divergence? There are many different possible explana
tions. One could be that the participants were drawn from a varied pool of
expertise, and individual differences in background, education, training,
experience, and decision-making roles could account for these variations.
Another possibility is that participants placed different interpretations on

the context of the problem. However, responses from individuals with similar backgrounds ranged across the full continuum. For example, the planners made comments that ranged across the continuum in spite of their similarities in training and decision-making roles. Therefore, it appears that individual cognitive style, irrespective of training or context, may be a more significant factor in explaining how people deal with conflicting cartographic information.

D Howe. Loution (SCAM:

In the broader context of cognitive psychology and human behavior, Jung (1983, 129) also questioned how individuals could interpret the same WISM material so differently? Jung concluded that people belonged to different psychological types and these types accounted for variations in individual decision-making processes. The basis of Jung's typology consists of four dichotomous variables: introversion/ extra version, intuition I sensation, thinking/ feeling, and judging/ perceiving. These variables combine to form 16 different psychological types. Jung (1983, 132) stated that there are never pure types and that all individuals exhibit varying degrees of all characteristics. A type is defined by the predominant mode of behavior. Readers may be more familiar with the extension of this work on personal ity types by Myers and Briggs (Myers 1962).

Keirsey and Bates (1984, 23) note that Jung proposed all of the dimen sions, but never fully developed the judging I perceiving dimension. Jung emphasized introversion/ extraversion as the predominant distinction between psychological types. Introverts emphasize the inner world of concepts and ideas, and extraverts emphasize the outer world of people Fig11re 3. l'v1ost Desirable Locatio11 for a House

and things (Myers 1980, 7). The other psychological functions describe by Map. how people perceive, come to conclusions, and come to closure. Intuition and sensing, according to Jung (1983, 144), are "the irrational functions" and relate to the perception of events or potential events. Intuition in Aspatial Spatial volves indirect perception, and sensing involves direct perception through combine

the five senses (sight, sound, smell, touch, and taste). Thinking and feeling are "the rational functions" and relate to how conclusions are made. polltlcal process

Thinking involves the use of logical processes aimed at an impersonal finding, and feeling involves processes based on subjective values (Myers Fig11re 4. Co11ti11u11111 of approaches for

1980, 3). The final dimension, judging and perceiving, involves preferences ha11d/i11g c01~f]icli11g cartograpl1ic approaches.

Aspatial Spatial for degrees of closure. Judging types prefer closure using the available data, and perceiving types defer closure desiring more data.

Returning to the idea of how participants made decisions using the
maps, it is possible to arrange several of the psychological functions along
a continuum from aspatial to spatial (Figure 5.). This provides a potential
Fix11r,· 5. C1>11ti111111111 of p.<_11cl10!.1girnl l!f/'''" a11d framework for explaining
the variations in approaches to conflicting
::pnf111/ dl'Ci,;1,111-111nki11s. information on maps. Common sense approaches align
with intuition and
feeling and generally appear to be aspatial, whereas modeling approaches
align with sensation and thinking and appear to be spatial. Spatial and
aspatial are defined solely on the basis of whether or not a map is neces
sary to the decision-making process. Furthermore, intuitive types prefer
approaches that are figurative and employ approximations; in contrast,
sensing types prefer approaches that are literal and employ detailed facts
(Kroeger 1992, 33). Common sense involves "unreflective opinions" (G. &
"Spatial and aspatial are defined C. Merriam 1987, 266), rather than careful evaluation of
facts. As noted
earlier, since common sense does not rely explicitly on site-specific criteria,
solely on the basis of whether or
the aspatial end of the continuum is defined as being largely intuitive. In
not a map is necessary to the
contrast, the spatial end involves the use of site-specific criteria for evalua
decisio11-111aki11g process."
tion and, therefore, is largely based on sensing.
It is important to note that these psychological dimensions describe a
continuum of approaches to decision-making behavior, and each is valid
in its own right. These define predominant decision-making preferences
and it is possible for people to shift style in some circumstances. The
predominant type of introversion I extra version and the functions of
perceiving/ judging do not appear to play a role in this context.
Variations in psychological types influence group dynamics in decision
making. The interaction of contrasting types has the potential for increas
ing both understanding and misunderstanding. This became apparent in
our focus group discussions where participants expressed a number of
contrasting opinions about the utility and necessity of the maps. In group
decisions, people cannot assume that everyone is perceiving and evaluat
ing the data in the same way and therefore will arrive at the same decision.
In the focus groups, we observed contrasts in psychological types; decision
makers exhibited both aspatial and spatial behaviors as described above.
The fact that the maps presented conflicting information may have rein
forced each participant's preexisting tendencies toward a psychological
"/11 group decisions, people type. Those that tended toward the aspatial end of the
continuum readily
cannot ass11111e that eueryone is discounted the validity of the maps and their importance

to decision
percei'ui ng and evaluating tile making. In contrast, those that tended toward the spatial end of the
data in tile same way and continuum looked for refinements to the models in order to improve the
maps. In this situation, the discussions were congenial, since no decision
therefore will arrize at the same
1
had to be made. However, the dynamics in groups that need to make a
decision. "
decision are often less congenial.
At the outset of this project, the application of the Jungian typology was
not anticipated, however, this typology provides a way to frame the
problem and provide some explanation. Contrasts in styles have a poten
tial for helping to understand how people process and manage conflicting
information in a cartographic context. In contexts where multiple decision
makers interact to solve a particular problem, in this case policy concern
ing groundwater vulnerability, diverse decision-making styles come into
play. The likely result of varied decision-making styles is multiple out
comes that are internally consistent to the individual decision makers be
cause each person frames the problem context according to her /his psy
chological type. Diverse results should serve as a reminder to those who
are cartographically oriented that not all users approach maps with the
same enthusiasm. This alludes to an issue of how we approach conflicting

information as a cartographic problem; the problem is not necessarily a
question of 'how maps work', but instead, 'how people work with maps'.
In this paper, results were presented from an experiment that investigated SUYfMARY AND
how decision-makers respond to conflicting cartographic information CONC lU SIOl\'S
using a series of conflicting groundwater vulnerability maps for one area.
Thirteen local decision-makers participated in the experiment, and input
\·Vas collected from them as to how they would deal with information
conflicts in a planning situation. The decision-makers then discussed the
maps and information conflicts in a group setting to examine how group
dynamics might influence the decision-making process. From this, a
continuum of strategies was developed that ranged from the aspatial
approaches, e.g., discard the maps and use common sense, to spatial
approaches, e.g., build a new spatial model that "works."
When the research was initiated, it was viewed as a problem of carto
graphic visual information processing and it was anticipated that the
participants would discuss methods for representing conflicting informa
tion. Hmvever, the issue turned very quickly away from the original maps "However, the
issue turned ztery

to one of how decisions are made using or, in some cases, not using the quickly away from the original
maps. This allowed the development of a continuum of approaches to the
maps to of how decisio11s are
011e
problem and a series of questions as to 'vhy such diverse results \·vere
made using or, i11 some cases,
obtained. It became apparent that very different internal processes were
11ot using the 111aps."
employed by the participants to evaluate the map data. The divergence of
responses suggested that something as fundamental as Jung's psychologi
cal types may play a role in how people evaluate map data.
The results from this study present several avenues for further research.
The first a\·enue deals with how to best represent information conflicts
and uncertainty between multiple maps. With growing reliance on GIS
modeling in environmental decision-making and issues of inter-model
stability, methods need to be developed to spatially represent agreement
and disagreement between models. MacEachren's (1994) suggestions
provide a starting point for such an investigation. The second avenue
deals with the role of individual preferences in decision-making and the
influence of psychological type on how people work with maps. It is
apparent from this research that there are diverse ways in ·which individu
als perceive and understand information and that this may impact the
decisions which the\· ultimate!\· make. Finallv, a third avenue deals with
how groups mal-.e d.ecisions u;ing maps. The' dynamics of individuals and "The role and
i111porta11ce of
groups in the decision-making process are markedly different, and little
multiple represe11tatio11s in
cartographic research has addressed this interaction.
visualizing spatial problems is
The role and importance of multiple representations in visualizing
well docu111e11ted, however, there
spatial problems is well documented, however, there needs to be a concern
with the quality of the information that these multiple ,·iews present. The 11eeds to be a
co11cem ·with the
potential for conflicting information in cartographic displays becomes quality of the
i11for111atio11 that
quite high when multiple data layers are combined in GIS models. The fact
these multiple viezus present."
that the s<1me data layers may produce markedly different maps \·\'hen
combined in different combinations or through different models is coming
to be a real concern for those who deal with environmental data and
decision-making. This concern becomes paramount in situations where

different representations may confuse or obscure understanding rather than illuminate it.

We thank the three anonymous re,·iewers for their comments and requests ACKt\O\ \'LEDGMENTS for clarification. \Ve also thank Dr. Rik Seefeldt from the Department of Psychology at the L"niversity of Wisconsin -River Falls for assistance with

issues pertaining to psychological typology. All omissions and interpreta tions however remain our responsibility.

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I
numerous health conditions and of equal size. These graphs show
reviews
diseases. lt may take a generation of not only the degree of dispersion,

medical geographers, epidemiolo but also whether variation is due to
gists and health services research relatively few outliers or is widely
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The Dartmouth Atlas of Health questions, hypotheses and intuitive text accompanying each map fo
Care. The Center for the Evaluative hunches derived from this single cuses on which hospital referral re
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cal School. John E. Wennberg, Prin Most of the maps in the atlas dis provides little interpretation of the
cipal Investigator and Series Editor, play variations in health-related data displayed. Some context is
Megan McAndrew Cooper, Editor. phenomena across hospital referral provided elsewhere. For example,
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diskettes); S350 (hardcover, with Medicare hospitalization data for "which surgical rate is right?" and
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gional supplements). ISBN: 1- ous regions based on patterns of delivery of health care services.
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information: AHA Services, Inc., referral regions were identified Hen/th Cnre is an impressive
P.O. Box 92683, Chicago IL 60675- through examining where residents achievement, it is not without its
2683, 1-800-AHA-2626). of each hospital service area were shortcomings. Foremost among referred for major card iovascu la r these is the failure to include a bib
Re1 1 iewed by: surgery and neurosurgery. A series liography would link the methods
Russell S. Kirby, Ph.D., M.S., F.A.C.E., of detail maps show the locations of used to the literature on health ser
Associate Professor each hospital referral region in rela vices research and medical geogra
Depnrt111e11t of0 /1stetrics n11d tion to major centers of health care phy, and would

connect each sta
Gy11ecology (p. 25-35). Ha\'ing defined the hos tistical map to its empirical re
Mi/u1n11kee Cli11icn/ Cn111p11s, pital referral regions, successive search base. To aid in interpreting
UW-Mndiso11 Medical Sc/100/ sections of the atlas provide detail the maps \•:ith data across hospital
ed maps of acute care hospital re referral regions, a map showing
Every now and again a revolution sources and expenditures; hospital populations in each region would
ary approach to the processing and capacity, utilization, and outcomes; have been useful. Most of the dis
presentation of information ap health manpm.ver (physicians); di ease and treatment data are derived
pears, changing forever the way in agnosis and surgical treatment of from the Medicare population,
which the subject matter is viewed common medical conditions. which is defined primarily by age
and understood. The publication of There are also sections on and almost universal participation.
The Dnrt111011th Atlas of Hen/th Cnrc benchmarking, tabulations of much Researchers interested in patterns
may be an example of one of these of the data presented in the detailed of health care use among individu
events. In this slender volume (this maps, and the strategies and meth als under the age of 65 will be dis
reviewer examined the soft-cover ods used to develop the information appointed in the minor omission of
academic \·ersion of the atlas; hard and maps in the atlas. Most of the maps for more than 80% of the
CO\'er and electronic versions are maps consist of two facing pages, population. There are no maps
also available), the project team has with a standard layout that in showing patterns of injury or dis
assembled numerous maps and vi cludes a national 1~ap •.vith five in ability, sur\'ival follm.ving specific
gnettes representing statistical sets (San Francisco, Chicago, De treatments, obstetrical procedures
analyses of literally tens of millions troit, Washington-Baltimore, and and outcomes, or spatial variation
of health care services, patients, New York) \·vi th the distribution of in propensity to use inpatient facili
and clinicians in a highly readable values for the phenomenon of inter ties among the non-elderly. This

and interpretable format. The atlas est grouped roughly into quintiles. omission is perhaps understand
portrays the current macro-scale Each discussion also includes a able, given the lack of standardized,
spatial structure of the American distribution graph showing the nationwide population-based data
health care delivery system and dis phenomenon of interest as the verti b,1scs on hospital services other
plays patterns of disease preva cal axis and each hospital referral than Medicare, yet there are some
lence and treatment modalities for region represented as a filled circle datasets (national vital statistics

data, for example) which, although British imperial expansion in South of language and religion, social and
not whollv satisfactorv for analvses Asia. Considering recent studies on economic information, mineralogi
at this scaie, could supply data' the duplicitous role of geography in cal, botanical and zoological inven
\·vi th \·vhich to address some of European colonialism, it is not sur tories were predicated on the notion
these issues. The section on prising that Matthew Edney has that the surveyor could achieve a
benchmarking, while interesting, is closely investigated cartography in complete scientific understanding
far too abbreviated to be of use to British India. His contribution to of the landscape through this im
those interested in applying these the study of geography and empire plicitly cartographic and visual
techniques to the planning or is to present a sophisticated expla framework. According to Edney, ob
evaluation of regionill health care nation of mapmaking in British In servation and reason provided a
delivervsvstems. dia as a cultural, political, and his "powerful rhetoric of vision, em
These criticisms notwithstand torical product. Rather than a ubiq piricism, and presence" that but
ing, this atlas is a monumental ac uitous tool for empire, mapmaking tressed the scientific authority of
complishment which all practitio in the divided bureaucratic context imperial cartography (p.75). This
ners of health services research or was confined, often contradictory, cultural and ideological expecta
medical geography will wish to ex and incomplete. The author's nar tion that science must

be a rational
amine. Geography and map librar rative brilliantly synthesizes text process of compilation set the
ies should definitelv add this vol and context-map and motive- to ground\"-'Ork for the construction of
ume to their holdings, and will find tell a story of how the British map what Edney names as the geo
the hardcover edition consulted fre ping of India was contingent on a graphical "archive." Th.is "archive"
quently. Spatial data repositories variety of competing practical and includes representations, images,
should consider acquiring the CD ideological interests. Throughout narratives, and maps assembled
ROMs \Vi th the detailed data used this book, Edney deploys a rigorous and ordered that underpinned the
to generate the maps and graphics theoretical analysis on voluminous larger cultural process of colonial
included in the atlas. archival material to illustrate the ism.
This reviewer looks forward to central theme in Mnppi11s n11 E111pire: Scientific expectation also
the linkage of spatial analyses of A tension between Enlightenment framed the conceptual scope, pro
variiltions in health care deliverv to epistemological ideals of obsen·a cess and practice of colonial
decision-making by health care or tion and notions of representation mapmaking. Edney illustrates how
gani,a1tions, insurers, policy-mak were manifested through the actual the British understood vision and
ers and health services researchers, practices of survey and mapmaking observation in the scientific pro
and to expanded and updated fu "on the spot." Moreover, this carto cess. He argues that scientific inves
ture editions of this iltlas. graphic project based on intersect tigation was an exercise of power,
ing ideologies of colonialism and explains how the picturesque land
contradictorv ideals of science was scape aesthetic fashioned images of
mediated through colonial institu South Asia, and outlines how an
Mapping an Empire: The Geo tional negotiation and historical implicit cartographic framework
graphical Construction of British contingencies. structured geographical narratives
India, 1765-1843. Matthew H. The first section of Mappi11g n11 that contributed to the colonial
Edney. Chicago: CniversityofChi Empire, Edney draws from many archive. Emphasizing the

cultural
cago Press, 1997. Pp. xv+ 340, empirical examples to outline how and scientific influence of the carto
maps, references, biographical overarching Enlightenment ideas graphic perspective on more gen
notes, and index. $35.00 hardback and cultural expectations of "sci eral imperial information gathering
(ISB\l 0-226-18487-0). ence" influenced colonial practices activities, the author observes that
of mapmaking in the early years of "geographical observation implic
Reuiewcd by: British colonialism in South Asia. itly constructs new knowledge
Dr. W. Elizabeth f epso11 For example, the reports by "peripa based on the spatial and distribu
Dcpnrf11H'11t ofG eosrnplly tetic officers" surveying the Indian tion of phenomena, in which re
U11il1crsif!f of Cnl(tim1in nt Los A11seles landscape, such as Colin spect is firmly rooted in the map
Mackenzie's Sun·ey of Mysore and mapmaking, but observation
Enlightenment ideals, brute colo (1800-1801) and Francis itself is part of the larger knowledge
nial realities, and practical bureau Buchanan's Bengal survey (1807- discourses constituted by texts,
cratic negotiations collide in Mat 1814), attempted to achieve the maps, images, and statistical cen
thew Ednev' s historv of carto ideal of taking a complete inven suses" (p. 46). Through his broad
graphic pr~ctice du~ing late eigh tory. Survey of town locations, land overview of British cartographic
teenth and early nineteenth-century resources, and rivers, descriptions history in South Asia, Edney sup-

ports the notion that knowledge is framework and set in motion a geo Mackenzie's attempt at a systematic
intimately linked to power, but he graphical certitude that fostered, topographical survey, William
grounds this perspective on de ironically, practical material and Lambton's trigonometrical survey
tailed discussion based on the spe intellectual anarchy that became of India's eastern coast, and the in
cifics of British colonialism in South characteristic of British mapmaking ability of the government to create a
Asia. in India. single cartographic institution to
Edney connects these scientific The scientific and cultural ideals coordinate a

comprehensive survey
ideals to imperial realities and of cartography also met financial of southern India. These enlighten
moves his analysis from how geo limits, logistical constraints, institu ing examples illustrate how,
graphical perspectives fashioned tional inertia, and stochastic colo through detailed archival research,
images of South Asia to how sur nial realities. Conditions of the co one can empirically link internal
veys and cartography were used to lonial administration circum administrative conflicts and finan
"discipline" the Indian landscape. scribed all efforts to coherently map cial constraints of cartographic
Statistical, astronomical, and route India. The divided information practice to contested ideas in carto
surveys formed the backbone of co management system, patronage, hi graphic practice. Edney shatters the
lonial cartographic knowledge. erarchical decision-making appara image of the ubiquitous colonial
However, for eighteenth-century co tus, separation between govern state unified in purpose, perspec
lonial administrators, the potential mental departments, lack of re tive, and practice. With sound em
of the survey was only realized sources, and expanding need for pirical evidence, he shows that cul
when it \vas incorporated into a geographical information all con tural expectation of "science" and
larger knowledge base, reconciled tributed to a chaotic practice of social realties within the British co
with multiple geographical sources, mapmaking. Moreover, these struc lonial system had a profound effect
and systematically reconstituted in tural constraints exacerbated the on the construction of geographical
a single encyclopedic cartographic tension between surveyors and knowledge.
archive. Yet, according to scientific Company cartographers in London The territorial expansion of Brit
expectations, the information was offices who eventually compiled the ish rule during the early nine
obtained only through the inchoate multifarious surveys into the single teenth-century demanded a unified
and imperfect vision of the sur vision of India. This permitted a cartographic project to deal with in
veyor. For the cartographic project "mini-hierarchy" of cartographic creased governmental

responsibili
to progress scientifically, the sur expertise to form. As Edney writes, ties. As a result, the British insti
veyors needed to find new, more "The surveyors worked at the tuted the Great Trigonometrical
comprehensive and "rational" boundary between the uppermost, Survey to support the publication of
methods of mapping the empire. British portion of the pyramid and the Atlas of India. In the final chap
What Edney emphasizes is that the lower foundations staffed by ters of Mappi11g an Empire, Edney
technological change to triangula Eurasians and Indians; the cartog writes a comprehensive history of
tion concommitantly transformed raphers functioned close to the pin this attempt to map "all of British
the practices of cartography in Brit nacle of the pyramid, collecting India." The Atlas of l11dia was in
ish India. The innovative trigono data and disseminating maps tended to unify disparate topo
metrical survey provided the "tech downward as necessary" (p. 162). graphical surveys based on a com
nological fix" to the acknowledged Edney focuses on the Madras mon "geometrical groundwork" of
imperfect, indirect, often ambiguous surveys (1790-1810) to best illus the Great Trigonometrical Survey of
route surveys, descriptive topogra trate the contingent character of India (GTS). This framework that
phies, and astronometrical observa British cartography as both duplici underpinned the Atlas of J11din
tions. Triangulation created a "rig tous in advancing British claims on "held the promise of a perfect geo
orously structured space" sup indigenous resources and divided graphical panopticon" to combine
ported by more accurately con in its practice. Lack of administra both geodetic science and genera 1
trolled measurements and provided tive unity was the prevailing char geography, thus reducing geo
the framework for more detailed acteristic in the mapping of Ma graphical data to a uniform refer
surveys to be integrated at large dras. Different motives and visions ence (p. 319). The complexity of this
scales. Trigonometrical surveys or of how to make an accurate and section is found within the empiri

dered cartographic space and disci comprehensive map in accordance cal examples of how the unified im
plined the Indian landscape to Eu with scientific and cultural expecta age of India developed historically
ropean science. This new expensive tions characterized this era of within the changing requirements
and time-consuming cartographic mapmaking. otable illustrations of colonial administration to rule
method reduced the Indian land of this cartographic anarchy are their newly acquired territory.
scape to a uniformed mathematical Edney's discussion of Colin Edney's discussion on the Great

Trigonometrical Survey and its in interested in the history of geo ing agencies include the National
stitutional development provides graphical thought. Biological Service, Oregon Depart
the reader with fundamental under ment offish and Wildlife (ODFW),
standing how the character of Brit Atlas of Oregon Wildlife: Distribu U.S. Environmental Protection
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mise. Shaughnessy, Eleanor P. Gaines, sity.
Archival synthesis and lucid and Manuela M. P. Huso. Corvallis: The Atlas is well-organized with
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The cartographic specialist will ap a comprehensive publication fea trial Breeding Vertebrates, (II)
preciate how Edney brilliantly inte turing information on Oregon's 426 Checklist of Wintering Birds, and
grates a profound understanding of native terrestrial vertebrate species (Ill) Winter Bird Distribution Maps.
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manpower, expense, and time lim cated to nearly each of the 441 spe tion cover types were then clustered
its-reinforces the broader theme cies. Each page contains a two into wildlife habitats based on
that cartography is a highly con color, 1 :4,300,000 scale, range map habitation by similar groups of spe
tested process within divided colo with supporting textual informa cies. A full-color 1 :750,000 scale Or
nial administration and limited re tion on Global Range, Habitat, Re egon Wildlife Habitat Map insert
sources. In Mappi11g an E111pire, production, Food Habits, Ecology, displays the habitat types and the
these logistical constraints are su and other relevant facts. Reference
perimposed on the cultural expecta to Order, Family, State and Federal
tions of science to show how the Status, Global and State Rank and
very fabric of geographical knowl Species Length are also listed. Each
edge-the map-is socially and po page contains a finely created line
litically constituted. For the colonial drawing of the featured animal. The

historian or cultural geographer in maps display the probable ranges
terested in questions of empire and where each wildlife species could
geography, Edney demystifies the be found, using shaded relief and
colonial state in the process of im county boundaries as spatial refer
perial expansion and brings into ence. Csuti states," The maps pre
focus the role of individuals and co sented here serve as a guide to habi
lonial institutions that have pro tats and general distribution of
found effects on how the British each species." The breadth and
proceeded to map India. Mapping depth of the information on wildlife
a11 Empire is both a monumental presented in this atlas is evidence
contribution to the history of British of a major collaborative effort. Many
colonialism and a necessary addi organizations are listed in the ac
tion to the libraries of geographers knowledgments. Thekeycontribut- Figure 1. Great Blue Heron

\'egetation CO\'er boundaries. This through each page uncomplicated.
mapping effort was part of the Or The complementary use of range
egon Wildlife Habitat Gap analysis maps, species drawings, and de
project. The habitat types were then scriptive text of micro-habitat con
overlaid with geographic units of ditions on a single page to commu
county, physiographic division, nicate probable places where a spe
and a network of .+4 l equal-sized cies could be found, work very well
Environmental Monitoring and As together. The species drawings re
sessment Program (E:vtAP) hexa ally help bring to life the atlas
gons covering the state. The geo pages. With a little, or for some a lot
graphic units contained informa of, imagination a reader can close
tion on presence or absence of the Fis11rc 3. Wc<ft>m Po11d Turtle their eyes and actually see a belted
species based on biological studies, kingfisher perched on a branch
museum records, and historical ob horn sheep and wild turkey -are over the quick-moving upper Des
servations dating back to the jour based on maps directly from the chutes River, watching and waiting
nals of Lewis and Clark (180-1- ODFW using no GIS modeling. for its next meal, or possibly a west
1806). This overlay process pro Other species ranges were modified ern pond turtle basking in the sun
\'ided resultant range maps that with elevation data, for example, on a mud bank above a small pond
were then reviewed by biologists, Townsend's mole was eliminated in the Willamette Valley.
and whose input was used in carto from the Coast Range. Some of the Interesting but less inspiring

graphic fine-tuning of the final rare species are represented as than the drawing and range map of
range maps. point symbols plotted at known lo the western pond turtle is the tie the
calities, the solitary sandpiper has atlas designers make between the
one location. The mule deer and atlas pages and the large-format
black-tailed deer are the onlv two Wildlife Habitat insert map. Stating
subspecies that were mapp~d in the which habitat type specifically is
Atlas. The authors handled this related to each species would have
problem by using two different val made a stronger connection. Even
ues of ink, darker for east of the though the data represented on the
Cascades Range (mule deer) and insert map is the basis for most of
figun• 2 . .\ 'ortha11 l~il't'r Oller lighter for west of the Cascades the range maps in the atlas, its pub
Range (black-tailed deer). lication with the Atlas appears to
Because of the characteristics of The authors of this atlas truly need be an afterthought. Unlike the maps
the species habitat or the a\·ailable to be congratulated on successfully on the Atlas pages, the insert map
data, several methodologies were publishing in an easy-to-access and could be characteriL:ed as a GIS
de,·eloped to crea tc selected species easy-to-use format a tremendous analysis product, produced on a
maps by either modifying the amount of geographic information medium-resolution inkjet plotter.
ranges derived from the modeling system data on a subject of great The 1:750,000 map design is lack
method mentioned '1bovc or by us interest to the Oregon public. In the ing in reference information, only
ing an entirely different approach. introduction, Csuti touches on this displaying the county boundaries
Some of the resultant maps from the basic function of the atlas, "While in addition to the habitat and veg
modeling received C1dditional atten the information gathered for GAP etation types. This map could have
tion by including major hydro Analysis is maintained as digital been a great addition in helping
graphic features for species, like the data in a geographic information meet one of the purposes the au
northern ri,·er otter, that are closely system, this atlas provides a larger thors state: "They (the maps) can
associated with water, \•,:ith the h\; audience 1,vith access to current direct you to areas where field stud
drograph~· printed in '1 higher - knowledgcaboutOregon'swild ies can determine if a

species has
\'<liue ink. Other species such as life." The OSU Press is already found the right combination of
the mallard duck or western pond gearing up for a second printing, habitat elements that enable it to es
turtle have a verv discontinuous attesting to the popularity of the tablish and maintain a popula
h<lbitat which is-denoted with a subject matter and the quick tion." Having little or no spatial ref
stippled area pattern for the range acceptance of the Atlas. erences diminishes its effective use
symbology. For marine birds a not The lc1yout and design of the as a location tool. Base GIS themes
to-scale buffer off the coast Ii ne was pages is clean and easy to read. The of major transportation, populated
gencrnted to displ<ly their h<lbitat. use of teal-colored ink for the title places and hydrography and eleva
Two introduced species-the big and subtitle makes navigating tion including shaded relief are

readily a\·ailable from the Oregon ized research and planning project. primary options offered to trans
StateSer\'ice Center forGISand The citizens of Oregon are \'ery form GIS data into a computer for
could ha\'e been added as a subtle lucky to be the recipient of this com mat for the production of high-qual
background to help with location. prehensive book. This atlas is an ity map products. Most cartography
As mentioned above there were educational tool that could lead to a labs adopted the use of graphic arts
actually several methods of compi greater awareness and sensitivity applications such as Adobe Illus
lation for the range maps. The need among Oregon's human popula trator or .\1acromedia Freehand'
for a legend on each map to help tion of the other inhabitants in their as digital tools for map production.
clarify these different compilation state. Even though these software pack
methods would have been an aid in Note: To view the range map of ages were fully capable of import
understanding the meaning of the the northern river otter you can go ing the points, lines, and polygons
different symbologies used. For ex to the website <http: I I from the GIS, these programs could

ample, the difference in the point bufo.geo.orst.edu I brc/ temp> and not take ad vantage of the useful at
symbols used in the solitary sand open 111·ottrnap.g~f. It is also possible tribute information that is main
piper (location from a scientific re to obtain a copy of the map image tained in the GIS database.
port) and the smooth general through the ftp site At the >JACIS XVI annual meet
boundaries used for the big horn <bufo.geo.orst.edu>, and log on as ing in San Antonio, many attendees
sheep (ODFW non-modeled range) anonymous. Change directories to were introduced to Avenza Soft
in comparison to the \'erv detailed pub, and "get" either the com ware, Inc.'s \1APublisher through
range boundaries gener~ted by the pressed tiff-format files, a workshop. The \1.APublisher
GIS modeling for the pronghorn are nrottmap.zip for PC users, or software de\'elopers addressed the
all very different in appearance but nrottmap.tif.gz for U:'-JIX users. issue of maintaining the \'aluable
there is no legend to clarify the dif attribute data and manipulation
ferences. By studying the introduc power of a GIS within Freehand or
I
tory text, an explanation of the Illustrator. \1APublisher version
cartographic techniques
,·ariation in symbology becomes 2.1incorporated38 filters designed
clearer. to import \'ector and raster data
I feel there is a need to point out with complete attribute tables intact
a minor issue in the citation regard GIS Data Made Manageable for for several major mapping software
ing the first delineation of the Cartographic Production file formats: ESRI Arc View shape,
physiographic di visions of Oregon. ESRI ARC I INFO generate, Maplnfo
The author writes that" The physi Lo11is Cross II f mid/mif, USGSDLGandSDTS,
ographic provinces of Oregon were Florida Rcsourn.?s & £1rl'iro11111e11tal AutoCAD DXF and geo-referenced
first delineated by Franklin and A11alysis Cotter TIFF and JPEG. In addition to basic
Dyrness (1973). .. ".Further search Florida Stnte U11iversity import capabilities, MA Publisher
ing reveals that Franklin and C220U U11iPersity Center enabled the user to change the na
Dyrness Vcgctntio11 ofO rego11 a11d Tallalinssee, FL 32306-2641 tive projection of the imported fi Jes
\ Va~/1i11gto11 (1969) cite Baldwin (850) 6.J.J-7361 and create supplementary graphic
(1964) and Baldwin in Geology of /cross@111a iler fsu .ed 11 databases.
Orcgo11 (1964) cites Dicken's Orcgo11 In April of 1995, the Florida Re
GeogrnpliJt ''. .. The geology is dis Background: sources & Environmental Analysis

cussed regionally following physi Center (FREAC) began the second
tf
ographic diYisions outlined by The widespread adoption of Geo Water Resources Atlas Florida
Dicken (1955) ... "I hope future edi graphic Information Systems (GIS) (WRAFJ. The editors and cartogra
tions address this point. technology by government agencies phers found that most of the data
Beyond the few weak points just and the private sector has made previously submitted by the au
con?red the atlas is a great success. \'ast quantities of digital data thors via hard-copy maps and
This atlas can be held up high as readily available to cartographers. tables were now maintained in ex
an e\.ample to many cartographers Initially, the high cost of hardware tensive GIS data sets. MA Publisher
contemplating assembling a state and software, Im...-to medium qual performed beyond expectation
\Vi Id life atlas. In a broader context, ity graphic output, and limited data when addressing these new data
this atlas serves GIS professionals sets made GIS less attractive to car formats. All GIS points, lines, and
as an excellent example of making tographers. The practice of scan polygons were imported with their
accessible to a large audience a ning and tracing output from GIS accompanying geographic accu
complex GIS database that was plots or importing vector line work racy and attribute tables without
originally generated for a special- into graphics programs were the error. However, some obstacles be-

yond MAPublisher's control were ter Management District's plan files. It is good practice to keep the
discovered while importing GIS ning/GIS division for a combined component parts of the shapefile
data. map. The files included a south (.shp, .shx, and .dbf) together as
Often the data were far too de Florida coastline (sf_shoreline), MAPublisher's built-in routines
tailed for use within the atlas. The 1996-water quality data by hydro search for them when importing.
level of precision and number of Iogic basin (complete_wq96) and Importing data through
points originally created in the GIS point locations of hazardous waste MAPublisher into Freehand was

caused difficulty when importing sites (hazard) (FIGURE 1). lluite straightforward. The import
files. Freehand Ts internal limit of Once imported into Arc View, the menu [Xtms! MAP11blisher/1111port ...]
32,000 points, for example, restrict coverages were converted to shape- offers the options in logical pull
ed larger ARC I INFO coverages down menus (FIGURE 2). In addi
from being imported. The coastline tion to file format and feature type,
files often required considerable the user must specify scale and
simplification as the base maps in page location for the element that is
the WRAF were much more general being imported. Within the pull
ized. MAPublisherenabled queries down fields the interface offers con
to be performed separating data venient browsing capabilities to lo
into useful groups. Attribute tables cate files. In order to maximize the
were created or edited within Free image, the default scale can be se
hand without having to edit in an lected and the resulting image will
external spreadsheet or database. fill the current page. for visual pur
Once the points, lines, and poly poses, this was found most useful
gons were imported, Free-hand's in the WRAF, as the graphics were
simplify [Xtrns!Cleanup!S.implifi;. ..] to be re-scaled to standard
routine was applied to remove un basemaps.
necessary points. The [A 111ou n t: To begin map construction, the
0.25] was realized as the optimal Figure 1. Ar<Virn• Dis11la.11 shapefile of south Florida's coast-
level of simplification, eliminating
£l
the most points while maintaining MAPublisher I mporl
path shape. MAPublisher's func
tionality provided a means of
streamlining the large GIS data set
and making final maps more attrac
tive and accurate.
r.'" Line
The follmving example illus
trates how ~APublisher assisted in Area
the construction of a graphic for the
WRAF. The production was done AutoCC..D DXF
Maplnfo mid/mif
using a 200 MHz Pentium Pro
USGS DLG
workstation "vi th 64MB RAM run Select
USGS SOTS
ning Microsoft Windows NYE 4.0

opera ting system software, ESRI
ArcV iew' 3.0a, Macromedia Free
hand 7.02, and A venza MAP ub
1
lisher 2.1 application programs. Scaling 1 to J Defaults
~---
Method: MapAnchor: Page Anchor:
X: r X:
Note: all kevboard shortcuts are r
Y: Y:
shown for the Windows I PC OS. Angle:
for Mac OS use simply substitute
J
the Command key for the Control Set
key(Ctrl).
For this example, three ARC/
INFO export files (.eOO) were ac
quired from the South Florida Wa- Figure 2. A.-IAP11/1/isl1cr Import Me1111

Number 29, Winter 1998 cartographic perspectives 55
line was imported first for registra down menus give access to 119 dif With the three shapefiles im
tion purposes and the establish ferent projections and 42 ellipsoids ported, it was determined that the
ment of default values (FIGURE3). contained in the filter (FIGURE 4). individual types of waste sites and
A search button complements the This feature was not used because water quality values for the final
filename field of the menu making it the files in the sample were all pro map needed to be extracted. When
jected to state plane coordinates. the attribute tables of the imported
Default values were chosen for the shape files were examined, legend
scale, page and map anchor fea categories were established. The
tures so the data would be bound to waste site coverage included three
the current page size. MAPublisher types of facilities: National Priority
creates and labels the new layer the List (Superfund), state-funded, and
filename of the import by default. landfills. Water quality was desig
This can be modified, but it was nated as either good, fair, poor or
found that for file management pur unknown. NlAPublisher'squery
poses, the default was preferred. features [Xtms I MAP Select by At
For the waste site layer, the shnpefile tribute ...] enable the userto create
format and paint feature type were selections using SQL type logic
specified. Once a file has been im statements (FIGURES). Because
ported, the scale in which it was this was the first query for the layer,

brought in becomes an option from lllitinl Selectio11 ... was chosen. AJ-
F1x111-r 3. lmpMtcd Sliap,filc the pull-down field. Selecting Same
as: sf_sl10reline commits the succes
unnecessary to commit the "8-dot- sive layers to the scale of that layer.
3" GIS filenames to memory. It is ef This step is repeated until all of the
ficient to have each layer projected shapefiles have been imported. It is
to the same coordinate system. recommended that all of the files be
Ho\·vever, re-projection of the data imported before any manipulation
is possible using MAPublisher's takes place. Premature resizing of
projection editor [Ktras I MAPub an image or individual layer can
lisher!Projectio11 Editor ...]. Pull- result in inaccuracy.
Fig11re 5. Attribute Selectio11 Me1111
£i
MAPublisher Projection Editor
though the user may enter an ex
Input Parameters pression directly, pull-down fields
simplify the task of creating an ex
Projection Custom Elpsoid Parallels
pression by using the attributes at
Projection: tached to the imported file. Once
the C11rre11t Expression: included the
E~soict
necessary equation, INSERT was
False Easting:
chosen transferring the formula to
False Northing: the recessed "expression area."
Central Meridian: Clicking OK completed the opera
Central Paralet: tion and returned the user to the
document with the specified items
selected (FIGURE 6).
For map construction, these
lected elements can be modified for
the cartographic presentation. It
was found that cloning and moving
the selection to a new layer allowed
the user to keep the original im
ported file unaltered for further que
ries and created a buffer for unfore
seen error (FIGURE 7). Once the
map was completed, the original
Figure 4. i'v!AP11l1/1sher Pn1iccticm Editor Mc1111 layer was deleted. In addition to the

verted plus sign shape into the

white box. Return to the document
leaving the £i11d A11d Replace menu
open by clicking anywhere in the
_,·_,
document window. Create a path
..
Io,9w shape (a closed path circle polygon
, was used for this example) on a
••P' ... ~-..-;~
,... .•.•.•. . ....:.,-,t_'d-@ ' new layer and copy it to the clip
..... ju::Jit board. Return to the Eind And Re
··•fo·f')!M"·~
••llillBx•·llllJl'.-f place menu and E.aste 111: the path
Figure 8. Find and Replace
shape for the To: field. Clicking on
Qiangeexecutes the replacement
(FIGURE 8). Note: any unlocked layer
co11tai11i11g the element to be cha11ged
will be affected. The registration point
for these e/e111e11ts is different i11 Free
f1g11n' 6. Text Elements Sl'ic<"ft'd . - ... ·· . . hand a11d Illustrator. 111 Freehand, it is
.....~
~
. ~..•..:. !,_. .. ,,,._, .
the upper-left corner, while Illustrator
....a.:•1."'•··
:::~::-~.·~;-~· uses the lower-left corner .
.... r!] ~'-t-Oi<t,_'
..... -:--.-r· • .L The above steps were repeated
··~'>$.:~ for both imported coverages resu !t
·..·.•. '•"'Q ..';.-." '_!.I. .;l>
••K•••1 ..-..i ing in eight layers of discrete data
(FIGURE 9). Once satisfied with the
. . . . · ~~ " ;: · ; · _ - '! > . i: l : _ l . f . ' . . import, polygons were filled, line
••• (~~-tl:f-': weights were changed and all of
, •·•· !·!!.I'"B"~ IZ'-•I
the layers grouped in preparation
• • · •l l l! l llF f· , - .. 1 ,, .; , . _ , , ,u J for refit to the atlas basemap. Due to
the close fit of the imported layers to
f1g11re 9 Co111pleted Import
•.
.,,...;.i-:. ~- . Water Quality and Hazardous Waste Sites
Figure 7. Ct>m>erkd Pn//1 Slrnpc,;

above technique, the efficient MAP
Lcge11d feature can separate ele
ments while assigning legend infor
mation and color based on at
tributes. watEf Quality
L~sl
MAPublisher imports GIS lines ouni.nolflln I,.0. -p-tl-o-ns-
directly as vector line work; how 0Poor
.... a~ -.
ever, point symbols are imported as -,:Air "'•a ~MAP...ti•she1 Import
- Good
text plus signs. Freehand 7 can eas .... a Guides
ily change the text to path shapes if Type of Site
• L.1ndfill
these steps are followed:
• SU1~funded
Select all of the text [Ctr/ A] and • tbbl)nal Pr1onty Ltsl
(Sup•nund)
convert to paths [Ctr/ Shift P]. Select
one of the converted path shapes
and copy it to the clipboard [Ctr/ CJ. .YJ 50MI
From the f.dit menu select £i11d A11d
Replace, then choose Graphics ...
[Cir/ Alt E]. The Attril111te: should be
set to Path Shape and C/1a11ge /11:
shou Id be set to Docu111c11t. Clicking
E.astc 111: under Fro111: inserts the con- Fis111·,· /0. Finni lvfnp

the standard basemap for the both a broad scope of current schol
WRAF, it was milnually resized. map library arship and the rich seventy-year
Title, scale and legend were added ulletin board history of geoscience research at
to the map to com plcte the graphic LSU. The DepartmentofGeography
(FIGURE 10). -~- & Anthropology collection reflects
the department's teaching and re
Conclusion: The Louisiana State University search focus by concentrating on
Cartographic Information Center acquiring maps depicting historical
While MAPublisherdid not directly and current Louisiana, the Gulf
affect the final look of the published /0'111 M. A11derso11 South, Latin America (with particu
maps in the atlas, it proved to be in Mnp Librnrin11 mrd Director lar emphasis on Mexico), East Asia,
valuable in converting the large Cnrtogrnphic l11for111ntio11 Ce11ter and Europe. The
specific research

quantities of data received for the Dept. ofG eogrnplry & A11tlrropology subjects supported by the collection
preparation of maps into a useable Lo11isin11a Stale U11ivcrsity include historical, cultural, eco
form. The final maps are primarily (504) 388-6247 nomic and physical geography, an
the result of cartographic design de jm1ders@ls11t 1 111.s11cc.ls11.ed 11 thropology, coastal and fluvial geo
cisions and the functions of the Ir ttp:llwww.cadgis.ls11.ed 11/cic! morphology, and geology. The re
graphics software, but MAPub gionally unique materials in the De
lishcr contributed significantly in The Louisiana State University Car partment collection include the
several \.vays. First, the final printed tographic Information Center (CIC) largest collection of original Louisi
maps are much more accurate in is a significant research resource in ana Sanborn Fire Insurance maps
the geographic location of features the Gulf South region. The Center's outside of Washington, DC the
due to the use of detailed GIS data. collection containing over 280,000 most complete set of historic U .5.
Second, the ability to take data sets maps and 81,000 aerial photo Coast& Geodetic Survey hydro
from multiple sources and re graphs makes it the largest map col graphic and topographic charts of
project them to a common projec lection in Louisiana and ranks it the Louisiana Gulf Coast housed in
tion allows the cartographer to among the larger academic map li the region, copies of historic maps
combine Javers and create custom braries in the Cnited States. In ad depicting Louisiana, and historic
maps. Finally, the timeand cost dition to the many maps and aerial aerial photographs of Louisiana
savings from using this alternative photographs, the collection in dating from the 1930s to the 1980s.
method were esscntia I to the cludes several regionally unique Among the unique materials cov
completion of the project in a timely cartographic resources which are ering areas outside of Louisiana is
manner. In many instances, the al available to the University commu a nearly complete set of original
ternative \·1.:ould have been to take nity, state and local agencies, busi U.S. Geological Survey topographic
printed output from theGIS, manu ness and private patrons, as well as quadrangle maps

printed before
ally trace the line work in a more patrons from around the world. 1940. In addition to maps of the
generalized form, scan the image, The Cartographic Information United States, the scope of geo
and retrace the line work in Free Cen-ter is funded and staffed by the science research at LSU O\'er the
hand. De-partmentofGeography&An past seventy years has brought ma
The utilization of MAPublisher thropology and is administratively terials depicting areas outside of
for this project and the techniques and physically separate from the the U.S. to the CIC. One example is
that were developed that led to the LSU Libraries. The CIC collection is a collection of over 3000 large scale
production of final printed color comprised of materials in two cate (1 :50,000) topographic maps of
maps has enhanced the ability of gories distinguished by ownership. Mexico "·hich provides over 96%
FREAC cartographers to produce The majority of the material in the coverage of the country. In addition
high-quality maps for publication collection was acquired by LSU to cartographic materials, the CIC
that are more accurate and more geoscience departments and is now houses the Dr. Robert C. West Latin
cost effective. More information owned by the Department of Ge American photograph collection
about MAPublishcr can be found ography & Anthropology. Addi containing over 6,000 black & white
on the website: bliJ2.;.il tionally, the CIC houses materials photographs and the Dr. Robert C.
w\vw.avenza.com. deposited by the Army Mapping West Slide Collection of over 10,000
Service and the Federal Depository slides from around the world.
Note: Full screc111 1 crsicms of the graph Library Program. Complementing the material
ics cn11 liefo1111d nt http:!! Although the Cartographic Infor amassed by the geoscience depart
128.186.177.25/nrt iclc.'11111! mation Center dates its founding to ments are the Army Mapping Ser
1960, the collection is the product of vice depository material from the

1950s and 1960s and the current In general, access to collection In addition to the daily map li
Government Printing Office Federal materials is primarily provided brary duties of patron service and
Depository Library Program through index maps for series, processing acquisitions, and the

(FDLP). The current FDPL material while other local finding aides are long term rehousing, retrospective
is located in the CIC as provided in consulted for individual maps. In cataloging, and preservation efforts,
the Separate Housing Agreement addition, the FDLP materials are the CIC staff is promoting aware
with the Regional Depository Li piece-level cataloged in the LSU Li ness of the Cartographic Informa
brary located in the LSU main li braries LOLA online catalog avail tion Center's collection through
brary. The combination of the able at http:/ /www.lib.lsu.edu/da tours, class presentations, and the
Army Mapping Service maps and tabases/ Iola.html. The Department World Wide Web. The Center's
the current FDLP arrivals provides of Geography & Anthropology col web site, www.cndgis.lsu .edrilcicl,
world wide medium scale topo lection materials are classified and was established in November 1995
graphic map coverage as well as filed using a locally developed sys to provide an overview of the hold
small to large scale map coverage of tem. An effort is underway to create ings. Although the majority of the
the United States. electronic databases to improve ac CIC holdings are not in an online
The Cartographic Information cess to the departmental material. catalog, the web site does list the
Center operates as a closed-stacks Completed databases are available CIC's significant special collections
library. The staff consists of one full locally for the Dr. Robert C. West and map series. When the map da
time professional assisted by a 1 Latin American Photograph Collec tabase for a specific area is comple
Full Time Equivalent student staff tion, as well as for the map hold ted, a paragraph describing the
who retrieve materials in response ings depicting Louisiana, :vlexico, map holdings is added to the web
to patron requests. In calendar year and the West Indies. The work on site. In addition to the descriptive
1997 the staff served 723 walk-in databases for the Dr. Robert C. West paragraphs and map series listings,
patrons who requested the retrieval Slide Collection, and for maps de a complete list of Louisiana historic
of a total of 1120 maps and 658 picting the United States is under aerial photographic coverage by

aerial photographs. Other patron \vay. The pre-1940 U.S. Geological parish and year is provided. A com
services include locating and rec Survey topographic quadrangle panion list of the CIC's holdings of
ommending digital data sources, map indexes were completed in aerial photomosaics by county and
geographic names research, and 1997. year depicting states other than
suggesting map vendors for private Because the material in the De Louisiana is also available. Finally,
purchases. These service requests partment of Geography & Anthro in order to illustrate the aerial pho
are received from walk-in patrons pology collection results from the tograph collection's scope, six rep
as well as via telephone, fax, mail, long history of LSC geoscience resentative aerial photos taken of
and e-mail. teaching and research, these materi the LSU campus from the 1930s to
In addition to providing daily als are often old, fragile, and require the 1980s are on the web site.
patron service, the Cartographic In preservation. In the past, original The Cartographic Information
formation Center staff is working to maps as well as copies of historic Center is located in room313
overcome problems common to maps were acquired. As a result, Howe-Russell Geoscience Complex,
many map libraries such as inad map materials in the collection LSU Campus, Baton Rouge, LA
equate space, the need for retro range from acetate-based film nega 70803 and is open to the public
spective cataloging, and preserva tives to fragile newsprint. Addition from8:00 a.m. to4:30 p.m. week
tion. As with other map libraries, ally, the bulk of the collection is days except University holidays.
the most pressing problem is inad housed in acidic containers. A re
equate space. The recently imple cently completed major preserva
mented weeding program designed tion project involved the transfer of
to remove duplicate out-of-scope historic U.S. Coast & Geodetic Sur The University of Iowa's Map
materials has barely kept pace with vey chart images from decomposing Collection
normal acquisitions. I\'aturally, acetate film negatives to stable
once the weeding is completed, nor mylar. Currently, an ongoing pres Mary Mel 11 roy
mal acquisitions \viii soon fill the ervation effort is underway to pre Gouem 111en t Pu/11 icntio11s Dcpart111e11t
recently available space. Addition vent daily wear on fragile maps by and Map Co/lectio11

ally, the acquisition of several large removing the maps from the general U11 iversi ty of Imm Libraries
research collections is pending un collection, rehousing them in pH Iowa City, IA 52242-1420
til additional space is available. It buffered containers, and placing (319)335-6247 or (319)335-5926
is hoped that additional space will them in less trafficked drawers. FAX: (319)335-5900
be acquired in the next three years. e-mail: 111ary-111ci11 roy@uiowa.cd11

The University of Iowa's Map Col from another library on campus, As you can see, the path to get
lection is the largest map collection and were able to purchase very ad ting the three machines was not
in Iowa, with more than 176,000 equate hardware to run Arc View 3. clear-cut, the results came to us in
sheet maps, as well as: copies of At that point, ArcV iew was relo phases, but the focus audience was
Sanborn fire insurance maps for cated from documents to the Map always the same. The administra
more than 1,000 Iowa cities from Collection to run on the new ma tors at the University Libraries are
the 1870's through the 1930's, either chine. Lastly, the Map Collection very supportive of having desktop
on paper or microfiche; Iowa obtained funding in late 1997 for mapping capabilities in the Map
county atlases for the period from two more desktop mapping work Collection. We had earlier made
the 1870's to the l 930's on micro stations through funds designated larger requests to university offices
film; over 600 Iowa plat books; and by the University's Office of the Pro for machinery and software for this
more than 100,000 aerial photos of \'OSt for instructional technology service and had always been re
Iowa counties, for various vears enhancements. Thus, by the end of jected. Thus, the piece-meal ap
since the 19-lO's. Although-stron 1998, the Map Collection will be proach was begun, and it paid off.
gest in Iowa and Mid'vvestern mate running desktop mapping software !'\ow time must be devoted to writ
rials, the collection is international on three networked machines in the ing policies, creating templates and
in scope and ~ubject coverage is Map Collection-92s reading room. pre-designed maps, hunting for tu

broad, excluding only geologic Because of the networked environ torials, grappling with levels of ser
maps, \vhich are collected at the Ge ment, the machines \-vill be linked to vice, finding staff time for this new
ology Library. each other and to a color laser service, and even figuring out how
Until ,1pproximately three years printer also purchased with the these large machines will fit into the
ago, there were no electronic carto technology enhancement funding. space available. But had we not
graphic products in the Map Col We can also explore additional li been willing to pre-define our audi
lection. When l joined the collection brary access for ArcView, since we ence and to advance our cause
in 1995 as half-time Map Librarian, have a library site license for gradually, I believe we would still
I brought with me an interest in ArcView. be waiting for our first desktop
desk-top mapping capabilities. I I have consciously used the mapping station in the Map Collec
had worked with federal docu phrase "desktop mapping," rather tion.
ments for manv \'ears and knew of than GIS in this article. At this
the wealth of r{u~eric data in the point, the goal of placing these ma
documents department. I had also chines and software in the Map
received training in 1992 on Collection is to give patrons access
NACISn~
ArcView as part of the ARL/GIS to higher-end mapping tools to [_
Literacy Project, and had been us which they otherwise \-Vould not
ing a stand-alone copy of ArcV iew have had access. The University of
in the Government Publications De Iowa has several true GIS labor~to
1997-98 NACIS OFFICERS AND
partment. Thus, the Map Collection ries on campus, located in aca
BOARD OF DIRECTORS
began acquiring cast-off CD-ROM demic departments, where students
d ri vcs from other 1i brary depart enrolled in genernl education/ in
NACIS OFFICERS:
ments and some fairly simple CD troductory classes can utilize vari
ROM mapping products (e.g., Map ous GIS software products. When
President: Patricia Gilmartin
Expert, Pro,imity). The Map Collec many of these students finish tak
tion still did not have anv hard ing these types of courses in geogra Department of Geography
ware to run higher end mapping phy, etc., they will not have access University of South Carolina
Columbia, SC 29208

programs, until a series of events to the departmental labs and to
(803)777-2989 fax: (803)777-4972
changed our situation for the better. electronic mapping capabilities in
First, the libr<H\' was "wired" in general. After speaking with profes e-mail: gilmartin-pat@sc.edu
the spring of 1997, enabling Map sors, students, and other librarians Term ex pi res: October 98
Collection machines to access the at ALA and elsewhere, it seemed
outside world. Then with the assis more useful to focus this beginning Vice President: Cynthia Brewer
tance of supportive library adminis service in the Map Collection to Department of Geography
trators, we utiliLed library develop ward those students (and faculty) Pennsylvania State Uni\'ersity
ment funds, money from the library- whose main needs can be met by University Park, PA 16802-5011
92s collection development general mapping programs such as (81-1)863-5072 fax: (814)863-7943
fund, some of my own map collec ArcV iew and Maptitude. email: cbrewer@essc.psu.edu
tion funds, plus" donated CPU Term expires: October 98

Secretary:JamesO. \!linton Jeremy Crampton (10 /99) NACIS Board Meeting
Map Library Department of Geography January 10, 1998
University of Tennessee George Mason University Lexington, KY
15 Hoskins Fairfax, VA 22030-4444
Knoxville, TN 37996-4066 (703)993-1217 Present: Andrews, Baruth, Brewer,
(423)97-l-4315 fax: (..+23)974-0560 email: jcrampto@gmu.edu Chu, Crampton, Gilmartin, Krejcie,
email: jim-minton@utk.edu Meacham, Minton, Nelson,
Term expires: October 99 Valerie Krejcie (10 I 98) Patterson, Peschel, Peterson, Stoll,
4020 Kirk St. and Thorne.
Treasurer: Sona Andrews Skokie, IL 60076
Univ. of Wisconsin-Milwaukee (847)677-1881 Michael Peterson recognized Pat
P.O. Box 413 email: t-krejcie@nwu.edu Gilmartin as the newly-elected
Milwaukee, WI 53201 President of ACIS and turned
(-ll-1)229-6057 fax: (414)229-2-181 James E. Meacham (10/99) over the meeting to her. Gilmartin
email: sona@csd.uwm.edu Department of Geography called the Board to order and wel
Term expires: October98 1251 L'niversity of Oregon comed the newly-elected officers:
Eugene, OR 97403-1251 Cynthia Brewer (Vice President),
Past-President: Michael Peterson (541)346-4870 fax: (541)346-2067 and James Minton (Secretary) and
Dept. of Geography /Geology jmeacham@oregon.uoregon.edu new Board members: Greg

Chu,
University of Nebraska-Omaha Elizabeth !'\elson, Jim Meacham,
Omaha, NE 68182 Elizabeth elson (10/99) and Jeremy Crampton. Continuing
(402)554-..+805 fax: (-102)55-l-3518 DepartmentofGeography Board members are:
Peterson,
email: geolib@cwis.unomaha.edu San Diego State University Thorne, Patterson, Krejcie, and
Term expires: October98 San Diego, CA 92182-4..+93 Stoll. Sona Andrews continues as
(6 19)594-8560 Treasurer. Gilmartin also thanked
NACIS EXECUTIVE OFFICERS: email: nelson16@mail.sdsu.edu Peterson for his hard work as
im
mediate past President, Craig
Christopher Baruth Thomas Patterson (10/98) Remington for his work as Secre
A GS Collection :\lational Park Service tary and the out-going Board mem
P.O. Box 399 Harpers Ferry Center bers for their service: Jim Anderson,
Milwaukee, WI 53201 P.O. Box 50, Publications Barbara Buttenfield, and Glen
(800)558-8993 fax: (-11-1)229-3624 Harpers Ferry, WV 25425 Pawclski.
NACISexecutiveofficers
FAX: (-l14) 229-3624 (304)535-6020 FAX: (304)535-6144 are: Sona Andrews, Susan
Peschel,
email: cmb@csd.uwm.edu email: t_patterson@nps.gov and Chris Baruth.
Sona Andrews Joseph Stoll (10/98) Cartographic Perspectives
L'niv. of Wisconsin-Milwaukee Dept. of Geog. & Planning CP Editorial Board reported on
P.O. Box 413 University of Akron the progress of searching for a new
\!lilwaukee, WI 53201 306 Carroll Hall editor. They recommended dis
(..+14)229-6057 fax: (414)229-2-181 Akron, OH 44325-5005 banding the three-year
rotation. A
email:sona@csd.uwm.edu (330)972-7621 guest editor would be responsible
email: joestoll@uakron.edu for one issue per year. Mike
Susan Peschel Peterson volunteered his name as
AGS Collection, P.O. Box 399 Kathryn Thorne (10/98) editor for a threevearterm effective
Milwaukee, WI 53201 Dept. of Geography & Geology Fall/Winter1998. With Peterson
(800)558-8993 fax: (-114)229-362-l Belknap Hall absent, the Board moved, seconded,
email:sqp@gml.lib.uwm.edu Mansfield University and discussed the appointment.
Mansfield, PA 16901 The vote was unanimous in sup
NACIS BOARD Of DIRECTORS: (717)662-4612 fax: (717)662-4612 port of Peterson to
assume the role
email: kthorne@mnsfld.edu of Editor.
Gregory H. Chu (10/99) FurtherCP Ne.,,vs: Jim Ackerman
Dept. of Geog. and Earth Science and Jeremy Crampton have already
Univ. of Wisconsin-La Crosse volunteered tobefutureguestedi
LaCrosse, 'NI 54601 tors. Joanne Perry will assume role
(608)785-6675 fa"<: (608)785-8332 of Book Review Editor with assis
email:chu@mail.uwlax.edu tance. Several names were sug-

gested for CP Editorial Board. Spring Board Meeting tinuity and e'\perience provided by
Buttenfield offered to send flyers to Peterson noted that the experience Chris, Sona, and Susan form a
encourage contributions to CP. Mo with a telephone conference call for strong infrastructure for ;\IACJS
tion was made, seconded, dis the Board meeting did not work which is invaluable to its officers,
cussed and appron?d by the Board. well. Gilmartin asked for suggested board members, and membership.
It was noted that CP is getting meeting sites. Atlanta, Chicago, Las After a well-attended and infor
very close to the page limits for con Vegas, and Da Il as were offered. mative annual meeting in Lexing
sideration of moving to Perfect Gilmartin will determine best site ton last fall, we are looking forward
Binding. This move \Nould cost an and notify members in early 1998. to getting together again this year in
additional $120.00 for every addi Milwaukee (October 7 -10). Dr.
tional four pages. New Business Cvnthia Brewer is Vice President
Rowles mentioned an idea of giving a~d Program Chair this year. If you
Meeting Site For Year 2000 plaques to students who assist in have questions or suggestions
An open discussion was held the annual conference. She also about the annual conference, you
proposing se,·eral potential sites for noted that AAG meets March 24th- can reach Cindy at
I\ ACIS 2000. Members are encour 29th, 1998. cbrewer@essc.psu.edu (or at the De
aged to propose host institutions partment of Geography, Penn State
and sites and offer them to the Sub111itted, University, University Park, PA
Board for consideration. Submitting f i111 MillfOll 16802). The usual Call for Papers
sites should consider a host depart NACISSccretnry and details of the meeting will be
ment such as geology, geography, mailed out later this spring, but it is
or GIS lab. Other issues to consider not too early to be thinking about
are hotel and air fare costs. Cities attending and participating in the
suggested were: Las Vegas, Cleve fro111 tlic N ACIS Prcside11t meeting.
land, Omaha, Kansas City, Port co11ti1111edfro111pnge1 This year's annual meeting will
land or Eugene, OR, San Diego, be preceded by a full-day sympo
KnoX\'ille, and Saratoga Springs, board. In reality, Mike had been an sium on the topic of "Maps and
l\Y. obvious choice all along, but it just Minds: A History of Cartography in
took a little time to convince him. Geography Education." Organized
Membership Committee Mike has done an outstanding job by James R. Ackerman of the
Issue was rabed concerning "regu for a number of years as chair of the Hermon Dunlap
Smith Center for

lar student," "subscription," and editorial board, and I have no doubt the History of Cartography at
"affiliate" memberships. Other is that he will continue his exemplary Chicago's :\lewberry Library, the
sues relating to membership: The work as editor. I would like to symposium will feature si" to eight
establishment of a Bulletin Board thank Mike for accepting this re invited speakers who will examine
(electronic), mailings to GIS, carto sponsibility and to remind every the de\'elopment and use of maps
special interest groups (such as one else that we can make his life in Western education and publica
URJSA). as-editor easier by sending him tions from the sixteenth through the
more good manuscripts to consider twentieth centuries. This sympo
WEB Committee for publication in CP. sium, which has been several years
Jeremy Crampton offered tochaira The next reason I can enjoy serv in the planning, constitutes a rare
Web(\\'\\'W-type)Committee.Such ing as president of \JACIS is the tre opportunity to reflect on the interac
a site would cost about 525 per mendous job done by our E"ecutive tion between cartographic publica
month. Crampton and the Board Officers at CW-Milwaukee: Chris tion and societies' educational pri
will explore this issue further. Baruth, Sona Andrews, and Susan orities. Look for details in the Call
Baruth offered to work '"'·ith the Peschel. ACIS members may get for Papers for the annual meeting
Committee and the possibility of tired of hearing this same refrain and on our website at
moving UWM's "LeMdo" to the from every president, but the reason www.nacis.org.
Web. Parts of the pages could in we sound like a broken record is be Yes, 1998 is a good year to be
clude: NACISorgani1ation/ offic cause these three make the presi president of NACIS, mainly be
er.
ers, annual meetings, parts of dent's job such a ... \veil, ... a~ cause of the admirable support pro
The Boa rd wi 11 re-assess at the sure. They maintain our records vided by current and past officers of
spring meeting. and mailing lists, collect dues, man the organization. I invite you to join
age many of the arrangements for ACIS (if you are not already a
the annual conference, and handle member), renew your membership
innumerable other details. The con- (if you are late with your dues), and

participate (if you have been lurk perryj@ccmail.orst.edu. Reviews of Material should be submitted in
ing in the wings). I can assure you maps and mapping software will digital form on 3.5" diskettes. Please
that you will enjoy the results. be solicited by the Editor of CP. Pro send a paper copy along with the
spective reviewers are invited to disk. Text documents processed
Pat Grl111nrti11 contact the Editor directly at with \1acintosh or Windows soft
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Word, are preferred.
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The North American Cartographic Information Society
(NACIS) was founded in 1980 in response to the need for a multidisciplinary organization to facilitate communication in the map information commu nity. Principal objectives of NACIS are:
§ to promote communication, coordination, and cooperation among the producers, disseminators, curators, and users of cartographic information;
§ to support and coordinate activities with other professional organizations and institutions involved with cartographic information;
§ to improve the use of cartographic materials through education and to promote graphicacy;
§ to promote and coordinate the acquisition, preservation, and automated retrieval of all types of cartographic material;
§ to influence government policy on cartographic information.
NACIS is a professional society open to specialists from private, academic, and government organizations throughout North America. The society provides an opportunity for Map Makers, Map Keepers, Map Users, Map Educators, and Map Distributors to exchange ideas, coordinate activities, and improve map materials and map use. Cartographic Perspectives, the organization's Journal, provides a mechanism to facilitate timely dissemina tion of cartographic information to this diverse constituency. It includes solicited feature articles, synopses of articles appearing in obscure or non cartographic publications, software reviews, news features, reports (confer ences, map exhibits, new map series, government policy, new degree pro

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